

Characterization and classification of groups of multiplicative type

SGA 3, Exposé X

English draft

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About this draft

This is an unofficial English translation of Exposé X, “Characterization and classification of groups of multiplicative type” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo10-8nov09.pdf` (44 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé X. Characterization and classification of groups of multiplicative type

by A. Grothendieck

1. Classification of isotrivial groups. Case of a base field

Recall (IX 1.1) that the group of multiplicative type H over the prescheme S is said to be *isotrivial* if there exists a finite surjective étale morphism $S' \rightarrow S$ such that $H' = H \times_S S'$ is diagonalizable. When S is connected, S' decomposes as a finite sum of connected components S'_i , and one may therefore (replacing S' by one of the S'_i if necessary) suppose S' connected. Finally, one knows that one may dominate S' by an S'_1 finite étale connected, which is *Galois*, i.e. a principal homogeneous bundle over S with group Γ_S , where Γ is an ordinary finite group (cf. SGA 1, V nos. 7 & 3 when S is locally noetherian, and EGA IV₄, 18.2.9 in the general case).⁽¹⁾ We suppose S' chosen in this way, and propose to determine the groups of multiplicative type H over S which are “trivialized” by S' , i.e. such that $H' = H \times_S S'$ is diagonalizable.⁽²⁾ By descent theory (cf. SGA 1, VIII 5.4), the category of these H is equivalent to the category of diagonalizable groups H' over S' , endowed with operations of Γ on H' compatible with the operations of Γ on S' . (N.B. Since the groups considered are affine over the base, the question of effectiveness of a descent datum is resolved in the affirmative, cf. SGA 1, VIII 2.1.) Now S' being connected, the contravariant functor

$$M \mapsto D_{S'}(M)$$

is an anti-equivalence of the category of ordinary commutative groups with the category of diagonalizable groups over S' (cf. VIII 1.6), a quasi-inverse functor being $H \mapsto \text{Hom}_{S'\text{-gr.}}(H, \mathbf{G}_{m,S'})$.⁽³⁾

Proposition 1.1. *Let S be a connected prescheme, S' a connected principal covering of S with (finite) group Γ . Then the category of groups of multiplicative type over S split⁽⁴⁾ by S' is anti-equivalent to the category of Γ -modules, i.e. of ordinary commutative groups M endowed with a homomorphism $\Gamma \rightarrow \text{Aut}_{\text{gr.}}(M)$.*

One concludes in the standard manner:

Corollary 1.2. *Let S be a connected prescheme, $\xi : \text{Spec}(\Omega) \rightarrow S$ a “geometric point” of S , i.e. a homomorphism into S from the spectrum of an algebraically closed field Ω ; consider the*

⁽⁰⁾Version xy of November 6, 2009: Addenda placed in Sect. 9, revised through 8.8.

⁽¹⁾**N.D.E.** More precisely, this follows from the “axiomatic conditions of a Galois theory”, cf. SGA 1, V no. 4 g); when S is locally noetherian, the verification of the axioms is made in loc. cit., nos. 7 & 3, in particular 3.7, which rests on SGA 1, I 10.9. This latter result is proved, without noetherian hypotheses, in EGA IV₄, 18.2.9.

⁽²⁾**N.D.E.** In what follows one shall say that an S -group of multiplicative type is “split” if it is “trivial” in the sense of IX 1.2, i.e. if it is a diagonalizable S -group.

⁽³⁾**N.D.E.** S has been corrected to S' .

⁽⁴⁾**N.D.E.** Here and in what follows, “splittés” has been replaced by “déployés”, “splitte” by “déplote”, etc.

fundamental group of S at ξ (cf. SGA 1 V, no. 7):

$$\pi = \pi_1(S, \xi).$$

Then the category of isotrivial groups of multiplicative type H over S is anti-equivalent to the category of “Galois modules” under π , i.e. of π -modules M such that the stabilizer in π of every point of M is an open subgroup.

In this correspondence, with the isotrivial group of multiplicative type H is associated the group $M = \text{Hom}_{\Omega\text{-gr.}}(H_\xi, \mathbf{G}_{m, \Omega})$, where H_ξ is the fiber of H at ξ ; this group is naturally endowed with operations of $\pi_1(S, \xi)$.

Remark 1.3. We shall see below (cf. 5.16) that if S is normal, or more generally geometrically unibranch, every group of multiplicative type and of finite type over S is necessarily isotrivial, so classification principle 1.2 applies to groups of multiplicative type and of finite type over S , which correspond to the Galois π -modules of finite type over $\mathbf{Z}$. For the moment, let us restrict to the following result:

Proposition 1.4. *Let k be a field, H a group of multiplicative type and of finite type over k ; then H is isotrivial, i.e. there exists a finite separable extension k' of k which splits H .*

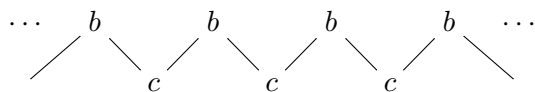
Consequently, by 1.2, if π is the topological Galois group of an algebraic closure Ω of k , the category of groups of multiplicative type and of finite type H over k is anti-equivalent to the category of Galois modules M under π of finite type as $\mathbf{Z}$ -modules.

It follows first from the fact that H is of finite type over k and from the “finite extension principle” (cf. EGA IV₃, 9.1.4) that there exists a finite extension k' of k which splits H . Recall the principle of the proof: by hypothesis there exists a diagonalizable group of finite type G over k , an S' faithfully flat over $S = \text{Spec}(k)$, and an isomorphism of S' -groups $H_{S'} \simeq G_{S'}$. Replacing S' by the residue field of a point of S' if necessary, one may suppose that S' is the spectrum of an extension K of k . The latter is the inductive limit of its subalgebras A_i of finite type, whence it follows easily (cf. EGA IV₃, 8.8.2.4) that u comes from an A_i -isomorphism $u_i : H_{A_i} \simeq G_{A_i}$ for sufficiently large i . By the Nullstellensatz there exists a quotient ring k' of A_i which is a finite extension of k . This therefore splits H .

Then k' is a radicial extension of a separable extension k'_s of k . By IX 5.4, the isomorphism $u' : H_{k'} \simeq G_{k'}$ comes from an isomorphism $H_{k'_s} \simeq G_{k'_s}$, which proves that k'_s splits H and establishes 1.4.

Remark 1.5. Statement 1.2 supplies in particular a characterization of isotrivial tori over S of relative dimension n : putting $\pi = \pi_1(S, \xi)$, they correspond to the classes (up to “equivalence”) of representations $\pi \rightarrow \text{GL}(n, \mathbf{Z})$ with kernel an open subgroup of π .

1.6. — Even when S is an algebraic curve, there may exist tori (of relative dimension 2) over S which are not locally isotrivial (and a fortiori not isotrivial); there may also exist locally trivial tori which are not isotrivial. (Note, however, that such phenomena can occur only if S is not normal, as already pointed out in 1.3.) Let, for example, S be an irreducible algebraic curve (over an algebraically closed field to fix ideas) having an ordinary double point a ; let S' be its normalization, and b, c the two points of S' over a . One then constructs a connected principal homogeneous bundle P over S with structural group $\mathbf{Z}$, by joining together infinitely many copies of S' according to the diagram



(N.B. This is a principal bundle in the sense of the étale topology.) Now one has a homomorphism

$$\varphi : \mathbf{Z} \rightarrow \text{GL}(2, \mathbf{Z}), \quad \varphi(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix},$$

which allows one to construct a torus T over S of relative dimension 2 from the trivial torus $\mathbf{G}_m^2$ over P and the descent datum on the latter deduced from φ . (N.B. One will note that the projection $P \rightarrow S$ is covering for the étale topology and a fortiori for the canonical topology of (Sch), and that the descent datum considered is necessarily effective, since $\mathbf{G}_{m,P}^2$ is affine over P .) It is not difficult to prove that T is not isotrivial in a neighborhood of $a^{(5)}$ (it is nevertheless trivial on $S - \{a\}$).

One finds a variant of this construction by taking for S a curve having two irreducible components S_1, S_2 meeting at two points a', a'' , which allows one to construct a connected and locally trivial principal homogeneous bundle P over S with group $\mathbf{Z}_S$, whence an associated torus T which is locally trivial but not isotrivial.

2. Infinitesimal variations of structure

⁽⁶⁾ Let us begin by recalling the following result (cf. SGA 1, I 8.3 in the case of S locally noetherian, EGA IV₄, 18.1.2 in general).

Recollection 2.0. *Let S be a prescheme, S_0 a subprescheme with the same underlying set. Then the functor*

$$X \mapsto X_0 = X \times_S S_0$$

is an equivalence between the category of preschemes étale over S and the analogous category over S_0 .

Proposition 2.1. *Let S be a prescheme, S_0 a subprescheme with the same underlying set (i.e. defined by a nil ideal $\mathcal{I}$). Then the functor*

$$H \mapsto H_0 = H \times_S S_0$$

from the category of group preschemes of multiplicative type over S to the analogous category over S_0 is fully faithful.

Moreover, it induces an equivalence between the category of quasi-isotrivial groups of multiplicative type over S and the analogous category over S_0 .

Let us first prove full faithfulness, i.e. that if H, G over S are of multiplicative type, then

$$\mathrm{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0)$$

is bijective. The question being local on S , one may suppose S affine; there then exists a faithfully flat and quasi-compact morphism $S' \rightarrow S$ splitting H and G . Let $S'' = S' \times_S S'$; denote by H', G' , resp. H'', G'' , the groups deduced from H, G by the base change $S' \rightarrow S$, resp. $S'' \rightarrow S$; define similarly S'_0, S''_0 , the latter being also isomorphic to $S'_0 \times_{S_0} S'_0$. One then finds a commutative diagram with exact rows:

$$\begin{array}{ccccc} \mathrm{Hom}_{S\text{-gr.}}(H, G) & \longrightarrow & \mathrm{Hom}_{S'\text{-gr.}}(H', G') & \rightrightarrows & \mathrm{Hom}_{S''\text{-gr.}}(H'', G'') \\ \downarrow u & & \downarrow u' & & \downarrow u'' \\ \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0) & \longrightarrow & \mathrm{Hom}_{S'_0\text{-gr.}}(H'_0, G'_0) & \rightrightarrows & \mathrm{Hom}_{S''_0\text{-gr.}}(H''_0, G''_0), \end{array}$$

so to prove that u is bijective, it suffices to prove that u' and u'' are so, which reduces us to the case where H and G are diagonalizable, hence of the form $D_S(M)$ and $D_S(N)$, where M, N are ordinary commutative groups. One shall therefore similarly have $H_0 = D_{S_0}(M)$,

⁽⁵⁾**N.D.E.** See 7.3 below.

⁽⁶⁾**N.D.E.** This “recollection” has been added, for later references.

$G_0 = D_{S_0}(N)$. One then has a commutative diagram⁽⁷⁾

$$\begin{array}{ccc} \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) & \xrightarrow{\sim} & \mathrm{Hom}_{S\text{-gr.}}(D_S(M), D_S(N)) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0}) & \xrightarrow{\sim} & \mathrm{Hom}_{S_0\text{-gr.}}(D_{S_0}(M), D_{S_0}(N)), \end{array}$$

where the horizontal arrows are isomorphisms by VIII 1.4, so one is reduced to proving that the homomorphism

$$(x) \quad \mathrm{Hom}_{S\text{-gr.}}(N_S, M_S) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(N_{S_0}, M_{S_0})$$

is bijective, i.e. to proving that the functor $M_S \mapsto M_{S_0}$ from constant commutative group preschemes over S to constant commutative group preschemes over S_0 is fully faithful. Now (x) is also identified with the natural map

$$\mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_S)) \longrightarrow \mathrm{Hom}_{\mathrm{gr.}}(N, \Gamma(M_{S_0}))$$

deduced from $\Gamma(M_S) \longrightarrow \Gamma(M_{S_0})$; now the latter map is evidently bijective (since $\Gamma(M_S)$ = the set of locally constant maps from S to M depends only on the underlying topological space of S), whence the desired conclusion.

To prove the second assertion of 2.1, it remains to see that every quasi-isotrivial group of multiplicative type H_0 over S_0 comes from a quasi-isotrivial group of multiplicative type H over S . To see this, let $S'_0 \longrightarrow S_0$ be a surjective étale morphism splitting H_0 .

One knows (cf. 2.0) that there exists an étale morphism $S' \longrightarrow S$ and an S_0 -isomorphism $S' \times_S S_0 \simeq S'_0$, so one may suppose that S'_0 comes from S' by reduction. Since H'_0 is diagonalizable, one sees immediately that it is isomorphic to the group deduced by base change $S'_0 \longrightarrow S'$ from a diagonalizable group H' over S' (N.B. if $H'_0 = D_{S'_0}(M)$, one takes $H' = D_{S'}(M)$). Put, as usual, $S'' = S' \times_S S'$, $S''' = S' \times_S S' \times_S S'$, and define similarly S''_0, S'''_0 , deduced from the preceding ones by base change $S_0 \longrightarrow S$ and also isomorphic to the fibered square resp. cube of S'_0 over S_0 . Using the full faithfulness already proved in the cases (S'', S''_0) and (S''', S'''_0) , one sees that the natural descent datum on H'_0 relative to $S'_0 \longrightarrow S_0$ (cf. IV 2.1) comes from a well-determined descent datum on H' relative to $S' \longrightarrow S$. This descent datum is effective since H' is affine over S' (SGA 1, VIII 2.1); there therefore exists an S -group H such that $H \times_S S' = H' = D_{S'}(M)$, and H is thus of quasi-isotrivial multiplicative type.

One then easily checks, now using the full faithfulness result for (S', S'_0) , that the given isomorphism between H'_0 and $H' \times_{S'} S'_0$ comes from an isomorphism between H_0 and $H \times_S S_0$. (For a more formal exposition of results of this type, see Giraud's article in preparation⁽⁸⁾ on descent theory.)

Corollary 2.2. *Let H be an S -group of multiplicative type, and $H_0 = H \times_S S_0$. For H to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial), it is necessary and sufficient that H_0 be so.*

The “necessary” part is trivial; the “sufficient” part has already been seen in the quasi-isotrivial case, since thanks to full faithfulness it suffices to know that every quasi-isotrivial group over S_0 lifts to a quasi-isotrivial group over S . The same argument works for “trivial”. For the “isotrivial” case one repeats the argument establishing the second assertion of 2.1, but taking $S'_0 \longrightarrow S_0$ étale surjective and finite. The “locally isotrivial” and “locally trivial” cases follow immediately from the “isotrivial” and “trivial” cases.

One may generalize 2.2 somewhat when $\mathcal{I}$ is nilpotent, by not supposing a priori H of multiplicative type:⁽⁹⁾

⁽⁷⁾**N.D.E.** The original has been corrected by interchanging M and N in the right-hand terms.

⁽⁸⁾Cf. J. Giraud, *Méthode de la descente*, Bull. Soc. Math. France, Mémoire 2 (1964).

⁽⁹⁾**N.D.E.** Recall that 2.3 is used to prove theorem IX 3.6 bis.

Corollary 2.3. *Suppose the ideal $\mathcal{I}$ defining S_0 locally nilpotent. Let H be a group prescheme over S , flat over S , and $H_0 = H \times_S S_0$. For H to be of quasi-isotrivial multiplicative type, it is necessary and sufficient that H_0 be so.*

Indeed, suppose that H_0 is of quasi-isotrivial multiplicative type, and let us prove that H is so. The question being local for the étale topology, and the category of preschemes étale over S being equivalent to the category of preschemes étale over S_0 by the functor $S' \mapsto S' \times_S S_0$ (cf. 2.0), one is reduced immediately to the case where H_0 is diagonalizable, hence isomorphic to a group $D_{S_0}(M)$. Let $G = D_S(M)$; one therefore has an isomorphism $u_0 : H_0 \xrightarrow{\sim} G_0$; I say that it comes from a unique homomorphism $u : H \rightarrow G$, which will therefore be an isomorphism since u_0 is so (H and G being flat over S and $\mathcal{I}$ locally nilpotent⁽¹⁰⁾), which will establish 2.3. Now one has (cf. VIII 1 (xxx))

$$\mathrm{Hom}_{S\text{-gr.}}(H, G) \simeq \mathrm{Hom}_{S\text{-gr.}}(M_S, \underline{\mathrm{Hom}}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})),$$

and the right-hand side is also identified with

$$\mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})),$$

so the homomorphism

$$(x) \quad \mathrm{Hom}_{S\text{-gr.}}(H, G) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, G_0)$$

is isomorphic to the homomorphism

$$\mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S})) \longrightarrow \mathrm{Hom}_{\mathrm{gr.}}(M, \mathrm{Hom}_{S_0\text{-gr.}}(G_0, \mathbf{G}_{m,S_0}))$$

deduced from the restriction homomorphism

$$(xx) \quad \mathrm{Hom}_{S\text{-gr.}}(H, \mathbf{G}_{m,S}) \longrightarrow \mathrm{Hom}_{S_0\text{-gr.}}(H_0, \mathbf{G}_{m,S_0}),$$

so to prove that (x) is bijective it suffices to prove that (xx) is so. The question is local on S , so one may suppose S affine. Now $\mathbf{G}_{m,S}$ being commutative and smooth over S , the situation is covered by IX 3.6, which completes the proof.

Corollary 2.4. *Let A be an artinian local ring with residue field k , $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$.*

(i)⁽¹¹⁾ *Let H be a flat S -group prescheme locally of finite type, such that $H_0 = H \times_S S_0$ is of multiplicative type. Then H is of multiplicative type, of finite type, and isotrivial. In particular, every S -group prescheme H of multiplicative type and of finite type is isotrivial.*

(ii) *The functor $H \mapsto H_0$ is an equivalence between the category of groups of multiplicative type of finite type over A and the analogous category over k .*

Indeed, let H be as in (i). Then H_0 is of multiplicative type and locally of finite type, hence of finite type, hence isotrivial by 1.4. Thus, by 2.3 and 2.2, H is of multiplicative type (hence of finite type) and isotrivial. Assertion (ii) then follows from 2.1.

Corollary 2.5. *Let $S' \rightarrow S$ be a faithfully flat and radicial morphism.*

(i) *The functor $H \mapsto H' = H \times_S S'$ from the category of groups of multiplicative type over S to the analogous category over S' is fully faithful.*

Moreover, it induces an equivalence between the subcategories consisting of quasi-isotrivial groups of multiplicative type.

(ii) *If H is of multiplicative type, for it to be quasi-isotrivial (resp. locally isotrivial, resp. isotrivial, resp. locally trivial, resp. trivial), it is necessary and sufficient that H' be so.*

⁽¹⁰⁾**N.D.E.** Since u_0 is an isomorphism, it suffices to see that for every $h \in H$ the morphism $\phi : \mathcal{O}_{G, u(h)} \rightarrow \mathcal{O}_{H, h}$ is bijective. It is so by reduction modulo $I = \mathcal{I}_s$ (where s is the image of h in S), so its cokernel C satisfies $C = IC$, whence $C = 0$ since I is nilpotent. Then, since $\mathcal{O}_{H, h}$ is flat over $\mathcal{O}_{S, s}$, the kernel K of ϕ also satisfies $K = IK$, whence $K = 0$.

⁽¹¹⁾**N.D.E.** The statement has been rewritten retaining only the nontrivial implication, in order to bring it out.

Indeed, let $S'' = S' \times_S S'$ and $S''' = S' \times_S S' \times_S S'$; then the hypothesis that $S' \rightarrow S$ is radicial implies that the diagonal immersions $S' \rightarrow S''$ and $S' \rightarrow S'''$ are surjective, so base change by either of these immersions is covered by 2.1 and 2.2. In view of the fact that $S' \rightarrow S$ is a morphism of effective descent for the fibered category of groups of multiplicative type over preschemes (since it is faithfully flat and quasi-compact⁽¹²⁾), our assertion follows formally from 2.1 and 2.2 (cf. for a formal argument the already cited article of Giraud).

3. Finite variations of structure: Complete base ring

Lemma 3.1. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$, G, H two S -group schemes, with G of multiplicative type and isotrivial, H affine over S , flat over S at the points of $H_0 = H \times_S S_0$, H_0 of quasi-isotrivial multiplicative type. Then the natural map*

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \operatorname{Hom}_{S_0\text{-gr.}}(G_0, H_0)$$

is bijective.

For every integer $n \geq 0$, let $S_n = \operatorname{Spec}(A/I^{n+1})$, and let G_n, H_n be the groups deduced from G, H by the base change $S_n \rightarrow S$. Since G/S is of isotrivial multiplicative type and H affine over S , by IX 7.1 the natural homomorphism

$$\operatorname{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \varprojlim_n \operatorname{Hom}_{S_n\text{-gr.}}(G_n, H_n)$$

is bijective. On the other hand, by 2.3 the H_n are of quasi-isotrivial multiplicative type, and by 2.1 the transition homomorphisms in the projective system $(\operatorname{Hom}_{S_n\text{-gr.}}(G_n, H_n))_n$ are isomorphisms, whence immediately 3.1.

Theorem 3.2. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/I)$. Then:*

(i) *The functor*

$$H \longmapsto H_0 = H \times_S S_0$$

is an equivalence between the category of isotrivial groups of multiplicative type over S and the analogous category over S_0 .

(ii) *Let H be an S -group of multiplicative type and of finite type; for H to be isotrivial, it is necessary and sufficient that H_0 be so.*

PROOF. For (i), one may either repeat the proof of 2.1 or use 1.2, using in either case the fact that the functor

$$S' \longmapsto S'_0 = S' \times_S S_0$$

from the category of finite étale schemes over S to the category of finite étale schemes over S_0 is an equivalence (SGA 1, I 8.4), which one may also state (reducing to the case of S connected, i.e. S_0 connected), choosing a geometric point ξ of S_0 , by saying that the canonical homomorphism

$$\pi_1(S_0, \xi) \longrightarrow \pi_1(S, \xi)$$

is an isomorphism.

Let us prove (ii), i.e. that if H_0 is isotrivial, H is so. By (i), there exists an isotrivial group of multiplicative type G over S and an S_0 -isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

⁽¹³⁾ Since H is of finite type, so are H_0 and G_0 ; thus by IX 2.1 b), the type of G at each point of S is an abelian group of finite type, and hence G is of finite type over S . On the other

⁽¹²⁾N.D.E. Specify this point...

⁽¹³⁾N.D.E. The original has been expanded in what follows.

hand, by 3.1, u_0 comes from a homomorphism of S -groups

$$u : G \longrightarrow H.$$

Finally, since G, H are of multiplicative type and of finite type over S , and since u_0 is an isomorphism, by IX 2.9, u is an isomorphism (taking account of the fact that every neighborhood of S_0 in S equals S). $\square$

Corollary 3.3. *Let A be a complete noetherian local ring with residue field k .*

- (i) *Every group of multiplicative type and of finite type over A is isotrivial.*
- (ii) *The functor $H \longmapsto H \otimes_A k = H_0$ is an equivalence between the categories of groups of multiplicative type and of finite type over A and over k .*

⁽¹³⁾ First, (i) follows from 3.2 (ii) and 1.4; then (ii) follows from 3.2 (i), in view of the fact that H is of finite type if and only if H_0 is so (cf. the proof of 3.2 (ii)).

Remark 3.3.1. One will note that 3.3 gives, by 1.4, a complete classification of groups of multiplicative type and of finite type over A in terms of the topological Galois group of an algebraic closure Ω of k .

Remarks 3.4. Under the hypotheses of 3.2 (i.e. A noetherian, separated and complete for the I -adic topology, but without any other hypothesis on A/I), it will follow from no. 5 that the functor $H \longmapsto H_0$ from the category of groups of multiplicative type and of finite type over S to the analogous category over S_0 is still fully faithful (without isotriviality hypotheses).⁽¹⁴⁾

However, it is not in general essentially surjective; in fact there may exist an S_0 -group H_0 of multiplicative type and of finite type, locally trivial if one wishes (but not isotrivial), which does not come by reduction from a group of multiplicative type H over S .

To see this, let us return to either of examples 1.6 of a nonisotrivial group of multiplicative type over a nonnormal curve. One may evidently take this curve affine, say S_0 , and suppose it embedded in affine space of dimension 2, hence defined by an ideal J in $k[X, Y]$. We shall take for A the completion of this ring for the J -preadic topology, so that A is a regular ring of dimension 2, a fortiori normal. We shall see in 5.16 that it follows that every group of multiplicative type and of finite type over $S = \text{Spec}(A)$ is isotrivial; hence H_0 , which is of finite type and nonisotrivial, does not come from a group of multiplicative type H over S (for H would necessarily be of finite type, and hence isotrivial).

Lemma 3.5. ⁽¹⁵⁾ *Let S be a prescheme, $u : G \longrightarrow H$ a homomorphism of S -group preschemes locally of finite presentation and flat over S , U the set of $s \in S$ such that the homomorphism induced on the fibers $u_s : G_s \longrightarrow H_s$ is flat (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite).*

Then U is open, and the restriction $u|_U : G|_U \longrightarrow H|_U$ is a flat morphism (resp. smooth, resp. unramified, resp. étale, resp. quasi-finite).

Indeed, let V be the set of points at which u is flat (resp...). One knows that V is open (cf. SGA 1, I to IV in the locally noetherian case, EGA IV in general⁽¹⁶⁾), and that for $x \in G$ over $s \in S$ one has $x \in V$ if and only if $u_s : G_s \longrightarrow H_s$ is flat (resp...) at x (same reference; in the flat, smooth, étale cases one uses here the flatness of G and H over S). Since u_s is a homomorphism of locally algebraic groups, this also means that u_s is flat (resp...) everywhere (Exp. VI_B, 1.3), i.e. $s \in U$. Thus U is the inverse image of the open subset V by the unit section of G , hence open, and $V = G|_U$, so $G|_U \longrightarrow H|_U$ is flat (resp...), which

⁽¹⁴⁾**N.D.E.** Specify this reference, possibly adding a corollary in no. 5...

⁽¹⁵⁾Cf. also VI_B 2.5 for more systematic developments of this nature.

⁽¹⁶⁾**N.D.E.** Specify these references: Since G, H are locally of finite presentation, u is locally of finite presentation (IV₁, 1.4.3 (v)); then one applies IV₃, 11.3.10 and 13.1.4 for flat and quasi-finite, IV₄, 17.4.1, 17.5.1 and 17.6.1 for unramified, smooth and étale; compare with VI_B 2.5.

completes the proof. (N.B. In the “unramified” or “quasi-finite” case, the flatness hypothesis on G, H is unnecessary.)

Lemma 3.6. *Let H be a commutative algebraic group over a field k admitting an open subgroup G of multiplicative type. Then the family of subschemes ${}_nH$ ($n > 0$) of H is schematically dense; in particular, if one has ${}_nH = {}_nG$ for every $n > 0$, then $H = G$.*

Here ${}_nH$ denotes the kernel of $n \cdot \text{id}_H$.⁽¹⁷⁾ Let $\bar{k}$ be an algebraic closure of k ; it suffices to show that the family $({}_nH_{\bar{k}})_{n>0}$ is schematically dense in $H_{\bar{k}}$, for then the family $({}_nH)_{n>0}$ will be so in H (cf. IX 4.5). Thus one may suppose k algebraically closed, and hence G of the form $D_k(M)$, M an ordinary commutative group of finite type. Let $M_0 = M / \text{Torsion}(M)$; then $T = D_k(M_0)$ is the largest torus contained in G , and H/T is finite, so $H(k)/T(k)$ is annihilated by an integer $\nu > 0$. One may find finitely many elements $g_i \in H(k)$ such that

$$H = \coprod_i g_i \cdot G,$$

and one shall have $g_i^\nu \in T(k)$. Since k is algebraically closed, $\nu \cdot \text{id}_T$ is surjective in $T(k) \simeq k^{*d}$; thus, replacing the g_i by $g_i t_i^{-1}$, where $t_i \in T(k)$ is such that $t_i^\nu = g_i^\nu$, one may suppose that $g_i^\nu = 1$. If then n is a multiple of ν , one shall have

$${}_nH \supseteq g_i \cdot {}_nG,$$

and as (for variable $n > 0$) the family of the ${}_nG$ is schematically dense in G by IX 4.7, conclusion 3.6 appears.

Theorem 3.7. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_0 = \text{Spec}(A/I)$, and let H be an affine S -group scheme of finite type, flat over S at the points of H_0 , such that $H_0 = H \times_S S_0$ is of multiplicative type and isotrivial.*

Then there exists an open and closed subgroup G of H , of isotrivial multiplicative type (and of finite type), such that $G_0 = H_0$.

By 3.2 (i), there exists an isotrivial group of multiplicative type G over S and an isomorphism

$$u_0 : G_0 \xrightarrow{\sim} H_0.$$

By 3.1, u_0 comes from a unique homomorphism of S -groups

$$u : G \longrightarrow H.$$

Using IX 6.6 one sees that u is a monomorphism (for if U is the set of $s \in S$ such that $u_s : G_s \longrightarrow H_s$ is a monomorphism, U is an open neighborhood of S_0 , hence identical to S , and $G|U \longrightarrow H|U$ is a monomorphism). By IX 2.5, u is even a closed immersion.

⁽¹⁸⁾ Thus G is of finite type, hence of finite presentation over S . Then, by lemma 3.5 in the “étale” case, one sees that there exists an open subset U which is a neighborhood of S_0 , hence identical to S , such that $G|U \longrightarrow H|U$ is étale; hence u is étale, hence an open immersion (since it is an étale monomorphism⁽¹⁹⁾), which completes the proof.

Corollary 3.8. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, $S = \text{Spec}(A)$, $S_0 = \text{Spec}(A/I)$, and let H be an S -group prescheme of finite type, affine and flat over S .*

For H to be of multiplicative type and isotrivial, it is necessary and sufficient that H_0 be so, and that H satisfy one of the following conditions a), b) (which are therefore equivalent under the preceding conditions):

⁽¹⁷⁾**N.D.E.** The following sentence has been added.

⁽¹⁸⁾**N.D.E.** The following sentence has been added.

⁽¹⁹⁾**N.D.E.** Cf. EGA IV₄, 17.9.1.

a) The fibers of H are of multiplicative type and of constant type on each connected component of S .

b) H is commutative and the ${}_nH$ ($n > 0$) are finite over S .

These latter conditions are also implied by the following:

c) The fibers of H are connected.

Of course, if H is of multiplicative type (and isotrivial), conditions a), b) are satisfied by IX 1.4.1 a) and 2.1 c), so only the “sufficient” part requires a proof. We shall use the subgroup G indicated in 3.7. When condition c) is satisfied, one evidently has $G = H$ and is finished.

In case b), one notes that the immersion $u : G \rightarrow H$ induces an open immersion

$${}_nu : {}_nG \rightarrow {}_nH$$

which induces an isomorphism $({}_nG)_0 \xrightarrow{\sim} ({}_nH)_0$; as ${}_nH$ is finite over S , this implies immediately that ${}_nu$ is an isomorphism (the complement of its image is finite over S and reduces to $\emptyset$). By 3.6 it follows that the morphisms induced on the fibers $u_s : G_s \rightarrow H_s$ are isomorphisms, hence u is surjective, hence an isomorphism.

Finally, in case a) one may suppose S connected,⁽²⁰⁾ and it follows that for every $s \in S$, $u_s : G_s \rightarrow H_s$ is a monomorphism of algebraic groups of multiplicative type and of the same type over $\kappa(s)$.⁽²¹⁾ I say that such a homomorphism is necessarily an isomorphism (which will again complete the proof⁽²²⁾). Indeed, one may suppose, extending the base field if necessary, that the two groups over $\kappa(s)$ are diagonalizable, and then this follows from VIII 3.2 b) and from the fact that a surjective homomorphism of isomorphic $\mathbf{Z}$ -modules of finite type $M \rightarrow N$ is necessarily bijective.

Corollary 3.9. *Let A be a noetherian ring endowed with an ideal I such that A is separated and complete for the I -adic topology, and let H be an S -group prescheme of finite type, affine and flat over S , with connected fibers.*

For H to be an isotrivial torus, it is necessary and sufficient that H_0 be so.

4. Case of an arbitrary base. Quasi-isotriviality theorem

Let A be a local ring. Recall that one says with Nagata that A is *henselian* if every algebra B finite over A is a product of local algebras B_i finite over A .

Recollection 4.0. ⁽²³⁾ *Let A be a henselian local ring, k its residue field, $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$, and ξ a geometric point of S_0 . Then the functor*

$$X \mapsto X_0 = X \times_S S_0$$

is an equivalence between the category of étale coverings over S and the analogous category over S_0 (cf. EGA IV₄, §18.5). Consequently (cf. SGA 1, V), one has $\pi_1(S_0, \xi) = \pi_1(S, \xi)$.

Suppose in addition A noetherian and denote by A' its completion, $S' = \operatorname{Spec}(A')$. Then A' is a complete noetherian local ring, hence henselian (loc. cit., 18.5.14), and the functor

$$X \mapsto X' = X \times_S S'$$

⁽²⁰⁾**N.D.E.** The original indicated: “i.e. S_0 connected”. Note that, as A is separated for the I -adic topology, S_0 meets every connected component of S (and as A is complete, the connected components of S_0 and of S are in bijection).

⁽²¹⁾**N.D.E.** Since G_s and H_s are of the same type for every $s \in S_0$, and hence for every s .

⁽²²⁾**N.D.E.** For u will again be a surjective open immersion.

⁽²³⁾**N.D.E.** This recollection has been expanded, and remark 4.0.1 added.

from the category of étale coverings over S to the analogous category over S' fits into a commutative diagram

$$\begin{array}{ccc} \text{Cov. ét.}(S) & \xrightarrow{\quad} & \text{Cov. ét.}(S') \\ & \searrow \simeq & \swarrow \simeq \\ & \text{Cov. ét.}(S_0) & \end{array}$$

and is therefore also an equivalence of categories, whence $\pi_1(S_0, \xi) = \pi_1(S', \xi)$.

Remark 4.0.1. Since S is connected (A being local), it follows from 1.2 that the category of isotrivial groups of multiplicative type over S is equivalent to the analogous category over S_0 (and also over S' if A is in addition noetherian).

Lemma 4.1. Let A be a henselian local ring with residue field k , $S = \text{Spec}(A)$, $S_0 = \text{Spec}(k)$.

(i) The functor

$$H \mapsto H_0 = H \times_S S_0$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S_0 .

(ii) If, in addition, A is noetherian, denoting by A' its completion and by $S' = \text{Spec}(A')$, the functor

$$H \mapsto H' = H \times_S S'$$

is an equivalence between the category of finite groups of multiplicative type over S and the analogous category over S' .

As in 4.0, the second assertion is a consequence of the first; let us prove the latter. One already knows that the functor considered is essentially surjective, since every group of multiplicative type H_0 over $S_0 = \text{Spec}(k)$, finite and hence of finite type over k , is isotrivial by 1.4, and hence comes from an isotrivial group of multiplicative type over S by 4.0.1.

It remains to prove full faithfulness, i.e. that for two finite groups of multiplicative type G, H over S the map below is bijective:

$$\text{Hom}_{S\text{-gr.}}(G, H) \longrightarrow \text{Hom}_{S_0\text{-gr.}}(G_0, H_0),$$

(24) i.e. denoting by F the S -functor $\underline{\text{Hom}}_{S\text{-gr.}}(G, H)$, that the natural map

$$\text{Hom}_S(S, F) \longrightarrow \text{Hom}_{S_0}(S_0, F_0)$$

induced by the base change $S_0 \longrightarrow S$ is bijective. For this, in view of recollection 4.0, it suffices to prove the

Lemma 4.2. Let G, H be two finite groups of multiplicative type over a prescheme S . Then $F = \underline{\text{Hom}}_{S\text{-gr.}}(G, H)$ is representable by a finite étale scheme over S .

(25) Let $f : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism such that G', H' are diagonalizable. It suffices to show that $F_{S'}$ is representable by a prescheme X' étale and finite (hence affine) over S' , for the descent datum on X' relative to f (cf. VIII 1.7.2) will then be effective (by SGA 1 VIII, 2.1), whence the existence of an S -prescheme X such that $X \times_S S' = X'$, representing F and étale and finite over S (cf. SGA 1, V 5.7 and EGA IV₄, 17.7.3 (ii)).

One may therefore suppose $G = D_S(M)$ and $H = D_S(N)$, where M, N are finite commutative groups (cf. VIII 2.1 c)). Then $K = \text{Hom}_{\text{gr.}}(N, M)$ is a finite abelian group, and by VIII 1.5 one has an isomorphism

$$\underline{\text{Hom}}_{S\text{-gr.}}(G, H) \simeq K_S,$$

(24) **N.D.E.** The following has been added.

(25) **N.D.E.** The original has been expanded in what follows.

which completes the proof of 4.2 and hence of 4.1.

Proposition 4.3.0. ⁽²⁶⁾ *Let S be a henselian local scheme, s its closed point, $g : X \rightarrow S$ a morphism locally of finite type, x an isolated point of the fiber X_s .*

(i) Then $\mathcal{O}_{X,x}$ is finite over $\mathcal{O}_{S,s}$. (In particular, if the residue extension $\kappa(x)/\kappa(s)$ is trivial, $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{X,x}$ is surjective, by Nakayama's lemma.)

(ii) If in addition g is separated, $X' = \text{Spec}(\mathcal{O}_{X,x})$ is an open and closed subscheme of X , i.e. one has a decomposition as a disjoint sum $X = X' \amalg X''$.

PROOF. By the local form of Zariski's main theorem found in [Pes66] or [Ray70, Ch. IV, Th. 1], x has an affine open neighborhood $U = \text{Spec}(B)$ of finite type and quasi-finite over $A = \mathcal{O}_{S,s}$, and there exists an open immersion $U \hookrightarrow Y = \text{Spec}(C)$, where C is an algebra finite over A . (N.B. To reach this conclusion one may also use Chevalley's semicontinuity theorem (EGA IV₃, 13.1.4), then the form of Zariski's main theorem given in loc. cit., 8.12.8.)

Since A is henselian, Y is the disjoint sum of local schemes $Y_1, \dots, Y_n$, each finite over S , and the points of Y over s are the closed points $y_1, \dots, y_n$. Thus $x = y_i$ for some i , and $\mathcal{O}_{X,x} = \mathcal{O}_{U,x} = C_i$ is finite over A . Moreover, $X' = C_i$ is an open subscheme of U , hence of X . Suppose in addition g separated. Then, since the morphism $X' \rightarrow S$ is finite (C_i being finite over A), the same is true of the immersion $X' \rightarrow X$ (cf. EGA II, 6.1.5 (v)), so X' is also closed in X . $\square$

Lemma 4.3. *Let A be a noetherian henselian local ring, A' its completion, $S = \text{Spec}(A)$, $S' = \text{Spec}(A')$, s the closed point of S , G, H two S -group preschemes, with G of multiplicative type and of finite type over S , H locally of finite type and separated over S , H_s of multiplicative type, and H flat over S at the points of H_s .*

Let G', H' be deduced from G, H by the base change $S' \rightarrow S$. Then the natural map below is bijective:

$$\text{Hom}_{S\text{-gr.}}(G, H) \xrightarrow{\sim} \text{Hom}_{S'\text{-gr.}}(G', H').$$

Since S' is faithfully flat and quasi-compact over S , one knows by SGA 1, VIII 5.2 that this map is a bijection of the left-hand side onto the subset of the right-hand side consisting of the $u' : G' \rightarrow H'$ such that the two inverse images $u'_1, u'_2 : G'' \rightarrow H''$ of u' over $S'' = S' \times_S S'$ (by the projections pr_1, pr_2 of S'' onto S') are equal.

Thus everything amounts to proving that for every homomorphism of S' -groups $u' : G' \rightarrow H'$ one has $u'_1 = u'_2$. By density theorem IX 4.8 it suffices to prove that u'_1, u'_2 coincide on ${}_n G''$ for every integer $n > 0$. (N.B. The density theorem is needed here essentially in a case where the base prescheme, here S'' , is not locally noetherian.)

This reduces us, replacing G by ${}_n G$, to the case where there exists an integer $n > 0$ such that $n \cdot \text{id}_G = 0$, hence where G is finite over S . Similarly, let ${}_n H$ be the kernel of the n -th power morphism ϕ_n in H . (N.B. We have not supposed H commutative, so ϕ_n (resp. ${}_n H$) is not necessarily a group homomorphism (resp. a subgroup of H).) Since ${}_n H$ is defined as the fibered product of $\phi_n : H \rightarrow H$ and the unit section $\varepsilon : S \rightarrow H$, its formation commutes with every base change $T \rightarrow S$, i.e. one has $({}_n H)_T = {}_n(H_T)$.

⁽²⁷⁾ Denote by a subscript m on the right the reductions modulo $\mathfrak{m}^{m+1}$, where $\mathfrak{m}$ is the maximal ideal of A . Then H_m is flat over S_m (since H is so over S at the points of H_0); thus by 2.4, since H_0 is of multiplicative type and of finite type, the same is true of H_m . Hence each $({}_n H)_m = {}_n(H_m)$ is a subgroup of H_m , of multiplicative type, finite and flat over S_m .

In particular, $({}_n H)_0 = {}_n(H_0)$ is finite over S_0 ; since S is local henselian and ${}_n H$ is (like H) locally of finite type and separated over S , by 4.3.0 the local rings of ${}_n H$ at the points of

⁽²⁶⁾**N.D.E.** This recollection has been added; it is used in the proof of 4.3 (the reference to EGA IV₄, 18.5.11 not being entirely satisfactory).

⁽²⁷⁾**N.D.E.** The original has been expanded in what follows, on the basis of indications of M. Raynaud.

$({}_nH)_0$ are finite over S , and one has a decomposition as a disjoint sum of open subsets

$$(*) \quad {}_nH = {}_nH^+ \coprod {}_nH^-,$$

where $Z = {}_nH^+$ is finite over S , and ${}_nH^-$ lies over $S - \{s\}$.

Note that for every finite S -scheme Y (such as G), every S -morphism $Y \rightarrow {}_nH$ factors through Z , and that formation of decomposition $(*)$ commutes with the base change $S' \rightarrow S$ (where $S' = \text{Spec}(A')$, A' the completion of A).

Then $Z' = Z \times_S S'$ is a finite scheme over S' , as is $P' = Z' \times_{S'} Z'$. Denote by ν the restriction to P' of multiplication in H' , and by σ the automorphism of P' interchanging the two factors. Since P' is finite over S' and H' separated and locally of finite type over S' , by EGA II 5.4.3 and IV₁ 1.1.3, $Y = \nu(P')$ is a closed subscheme of H' , universally closed and quasi-compact, hence of finite type, hence proper over S' . Moreover $Y \rightarrow S'$ has finite fibers (since $P' \rightarrow S'$ does). Thus, since S' is noetherian, the morphism $Y \rightarrow S'$ is finite (cf. EGA III, 4.4.2). Since $Z' \subseteq Y$ and $Z'_m = Y'_m$ for every m , one has $Z' = Y$ by lemma IX 5.0, hence Z' is a subgroup of H' . Similarly, the kernel $K = \text{Ker}(\nu, \nu \circ \sigma)$ is a closed subscheme of P' such that $K_m = P'_m$ for every m (since $Z'_m = {}_nH_m$ is commutative), so $K = P'$, i.e. Z' is a commutative subgroup of H' . On the other hand, since each reduction $Z'_m = {}_nH_m$ is flat over S_m , by the “local criterion of flatness” (cf. EGA 0III, 10.2.2 or [BAC], III §5, Example 1 and Th. 1), Z' is flat over S' . Since, in addition, $Z'_0 = {}_nH_0$ is a finite group of multiplicative type over S_0 , hence isotrivial by 1.4, Z' is of multiplicative type (and isotrivial) over S' by 3.8 b). Since $S' \rightarrow S$ is faithfully flat and quasi-compact, one deduces that multiplication $Z \times_S Z \rightarrow H$ factors through Z and makes Z a subgroup of H , finite over S and of multiplicative type (since Z' is so).

Finally, by the remarks made above concerning decomposition $(*)$, Z' is the “finite part” of ${}_nH'$, and therefore the morphism $u' : G' \rightarrow {}_nH'$ takes its values in Z' . Since Z is of multiplicative type and finite over S , as is $G = {}_nG$, by 4.1 u' comes from a $u : G \rightarrow Z$, and hence $u'_1 = u''_2$. This completes the proof of 4.3.

Lemma 4.4.0. ⁽²⁸⁾ *Let A be a ring, $S = \text{Spec}(A)$, H an S -group scheme of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), (A_i) a filtering family of subrings of A whose inductive limit is A , $S_i = \text{Spec}(A_i)$.*

Then there exists an index i and an S_i -group scheme H_i of multiplicative type and of finite type, quasi-isotrivial (resp. isotrivial), such that $H = H_i \times_{S_i} S$.

Theorem 4.4. *Let S be a prescheme, H an S -group prescheme affine and of finite presentation over S , and $s \in S$. Suppose:*

$\alpha)$ H is flat over S at the points of H_s .

$\beta)$ H_s is of multiplicative type.

Then there exists an étale morphism $S' \rightarrow S$, a point s' of S' over s such that the extension $\kappa(s')/\kappa(s)$ is trivial, and an open and closed subgroup G' of $H' = H \times_S S'$, of multiplicative type and of finite type, isotrivial, such that $G'_{s'} = H'_{s'}$.

⁽²⁹⁾ a) Denote temporarily by $T = \text{Spec}(\mathcal{O}_{S,s})$ and let us show that by “descent of conclusions” one may reduce to the case where $S = T$. Indeed, suppose one has found an étale morphism $g : T' \rightarrow T$, a point $s' \in T'$ and a subgroup G' of $H' = H_{T'}$ satisfying the conditions of the statement; replacing T' by an affine open neighborhood of s' , one may suppose T' of finite presentation over T .

Since G' is isotrivial, there exists a finite surjective étale morphism $f : \tilde{T} \rightarrow T'$ such that $\tilde{G} = G' \times_{T'} \tilde{T}$ is isomorphic to $D_{\tilde{T}}(M)$, where M is an abelian group of finite type. Since

⁽²⁸⁾**N.D.E.** This lemma has been added; it is used several times in what follows.

⁽²⁹⁾**N.D.E.** The following reductions have been expanded (the original indicated: “One reduces immediately to the case where S is the spectrum of a noetherian local ring A , and s is the closed point of S .”).

T' , H , and $\tilde{T}$, G' are of finite presentation over $T = \text{Spec}(\mathcal{O}_{S,s})$, and $\mathcal{O}_{S,s}$ is the inductive limit of its subalgebras $A_i = \mathcal{O}_S(S_i)$, where S_i ranges over the affine open neighborhoods of s in S , by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3) there exists an index i , S_i -preschemes (resp. an S_i -group prescheme) of finite presentation $S'_i, \tilde{S}_i$ (resp. G'_i), and morphisms $g_i : S'_i \rightarrow S_i$, $f_i : \tilde{S}_i \rightarrow S'_i$ (resp. a morphism of S'_i -group preschemes $u_i : G'_i \rightarrow H \times_S S'_i$), from which f, g (resp. $u : G' \rightarrow H'$) come by base change $T' \rightarrow S_i$. Moreover, taking i sufficiently large, g_i will be étale, f_i finite surjective étale, and u_i an open and closed immersion (cf. EGA IV, 8.10.5 and 17.7.8).

Then $\tilde{G}$ comes from the groups $\tilde{G}_i = G_i \times_{S_i} \tilde{S}_i$ and $D_{\tilde{S}_i}(M)$; thus by EGA IV₃, 8.8.2 (i) (and VI_B, 10.2), there exists an index $j \geq i$ such that $\tilde{G}_j \simeq D_{\tilde{S}_j}(M)$, so G_j is of isotrivial multiplicative type. Denote by s'_j the image of s' in S'_j . Then the étale morphism $S'_j \rightarrow S_j \hookrightarrow S$, the point s'_j and the open and closed subgroup G'_j of $H \times_S S'_j$ satisfy the conditions of the statement. This shows that one may suppose $S = \text{Spec}(A)$ local with closed point s . b) Then A is the inductive limit of local rings A_i which are localizations of $\mathbf{Z}$ -algebras of finite type; put $S_i = \text{Spec}(A_i)$. Let us show that hypotheses α), β) “descend” to some A_i . Since $H \rightarrow S$ is of finite presentation, the set of $x \in H$ such that H is flat over S at x is an open subset W , containing H_s by hypothesis, and hence containing a quasi-compact open subset V containing H_s (for H_s being affine, one may cover it by finitely many affine open subsets contained in W). Then the open immersion $\tau : V \hookrightarrow H$ is of finite presentation, so $V \rightarrow S$ is also so.

Thus by EGA IV₃, 8.8.2 (and Exp. VI_B, 10.2 and 10.3) there exists an index i , an S_i -prescheme V_i (resp. an S_i -group prescheme H_i) of finite presentation over S_i , and an S_i -morphism $\tau_i : V_i \rightarrow H_i$ from which V, H, τ come by base change $S \rightarrow S_i$; moreover, taking i sufficiently large, τ_i will be an open immersion and V_i flat over S_i , by EGA IV₃, 8.10.5 and 11.2.6. Denote by s_i the image of s in S_i ; since the open immersion $(V_i)_{s_i} \hookrightarrow (H_i)_{s_i}$ gives, by the base change $\kappa(s_i) \rightarrow \kappa(s)$, the equality $V_s = H_s$, one already has $(V_i)_{s_i} = (H_i)_{s_i}$, i.e. $(H_i)_{s_i} \subseteq V_i$, so H_i is flat over S_i at the points of $(H_i)_{s_i}$. Finally, $(H_i)_{s_i}$ is of multiplicative type, since $H_s = (H_i)_{s_i} \otimes_{\kappa(s_i)} \kappa(s)$ is so. Thus the triple (S_i, H_i, s_i) satisfies the hypotheses of 4.4, and if the desired assertion is satisfied for this triple, it will also be so, by base change, for (S, H, s) . This reduces us to the case where A is local and noetherian. Let us now distinguish two cases.

1°) A is local noetherian and henselian. Let $\hat{A}$ be its completion, $\hat{S} = \text{Spec}(\hat{A})$ and $\hat{H} = H \times_S \hat{S}$. Apply theorem 3.7: one finds a $\hat{S}$ -group $\hat{G}$ of multiplicative type, isotrivial and of finite type, and a homomorphism of $\hat{S}$ -groups

$$\hat{u} : \hat{G} \rightarrow \hat{H}$$

which is an open immersion and a closed immersion, such that $\hat{u}$ induces an isomorphism $\hat{u}_0 : \hat{G}_0 \xrightarrow{\sim} \hat{H}_0$.

By remark 4.0.1, the base change functor by $\hat{S} \rightarrow S$ induces an equivalence between the category of isotrivial groups of multiplicative type over S and over $\hat{S}$; in particular $\hat{G}$ “comes from” an S -group of multiplicative type G , isotrivial and of finite type. By 4.3, $\hat{u}$ comes from a homomorphism

$$u : G \rightarrow H;$$

moreover, u is an open and closed immersion and induces an isomorphism $u_0 : G_0 \rightarrow H_0$, since this is so after the faithfully flat quasi-compact base change $\hat{S} \rightarrow S$. This therefore proves 4.4 in this case (taking of course, in the conclusion of 4.4, $S' = S$ and $s' = s$).

2°) A is local noetherian. Reduction to case 1°) is immediate by applying 1°) to the “henselization” A^h of A . More precisely, one sees easily (using SGA 1, I §5⁽³⁰⁾) that the local rings $\mathcal{O}_{S',s'}$ of étale S -preschemes S' endowed with a point s' over s such that $\kappa(s')/\kappa(s)$ is trivial form a filtering inductive system whose inductive limit is a noetherian henselian local

⁽³⁰⁾Or EGA IV₄, §18.6.

ring A^h (called the “henselization” of A); for details of this construction (due to Nagata in the normal case), cf. SGA 4, VIII §4.⁽³⁰⁾ The sorites of EGA IV₃, §8 then allow one, as in part a) of the proof, to deduce from a result known over the inductive limit A^h of the $\mathcal{O}_{S',s'}$ an analogous result over one of the (S', s') , which proves 4.4.

Corollary 4.5. *Let S be a prescheme, H an S -group prescheme of multiplicative type and of finite type. Then H is quasi-isotrivial, i.e. is split by a surjective étale morphism $S' \rightarrow S$.*

Indeed, let $s \in S$. By 4.4, there exists an étale morphism $S' \rightarrow S$, an $s' \in S'$ over s such that $\kappa(s) = \kappa(s')$, and a subgroup G' of H' , of isotrivial multiplicative type and of finite type, such that $G'_{s'} = H'_{s'}$. Since G', H' are of multiplicative type and of finite type, by IX 2.9 there exists an open neighborhood U' of s' such that $G'|U' = H'|U'$.

⁽³¹⁾ Suppose in addition S local henselian with closed point s ; then by EGA IV₄, 18.5.11 b) there exists a section σ of $S' \rightarrow S$ such that $\sigma(s) = s'$. (N.B. One may see directly that $B = \mathcal{O}_{S',s'}$ equals $A = \mathcal{O}_{S,s}$ as follows: by 4.3.0 (i), one has $B \simeq A/I$, and since B is an A -algebra of finite presentation and flat, I is an ideal of finite type (cf. EGA IV₁, 1.4.7), and $I = \text{Im}$ (where $\mathfrak{m}$ is the maximal ideal of A), whence $I = 0$.) Thus H is already isotrivial. One therefore obtains:

Corollary 4.6. *Let A be a henselian local ring, k its residue field, and π the topological Galois group of an algebraic closure of k .*

- (i) *Every group of multiplicative type and of finite type over $S = \text{Spec}(A)$ is isotrivial.*
- (ii) *The category of these groups over S is equivalent (via the functor $H \mapsto H \otimes_A k$) to the analogous category over $S_0 = \text{Spec}(k)$; it is therefore anti-equivalent, by 1.4, to the category of Galois modules under π of finite type as $\mathbf{Z}$ -modules.*

Corollary 4.7. *Under the conditions of 4.4, suppose in addition one of the following conditions satisfied:*

- a) *For every generization t of s , H_t is of multiplicative type and of the same type as H_s .*
 - b) *H is commutative and the ${}_n H$ ($n > 0$) are finite over S in a neighborhood of s .*
 - c) *For every generization t of s , the fiber H_t is connected.*
- Then there exists an open neighborhood U of s such that $H|U$ is of multiplicative type.*

Taking account of the fact that an étale morphism is open, one is reduced to proving that there exists (with the notation of the conclusion of 4.4) an open neighborhood U' of s' such that $G'|U' = H'|U'$. Put $S'' = \text{Spec}(\mathcal{O}_{S',s'})$; as G', H' are of finite presentation over S' , it suffices to show, by EGA IV₃, 8.8.2, that $G'' = H''$. One may therefore suppose $S = S''$, and then hypotheses (a), (b), (c) become those of the lemma below, whose proof is the same as that of 3.8:

Lemma 4.7.1. *Let S be a local scheme, s its closed point, H an S -group scheme of finite type, G an open and closed subgroup of H , of multiplicative type, such that $G_s = H_s$. Suppose in addition one of the following conditions satisfied:*

- a) *The fibers of H are of multiplicative type and of the same type as H_s .*
 - b) *H is commutative and the ${}_n H$ ($n > 0$) are finite over S .*
 - c) *The fibers of H are connected.*
- Then $G = H$.*

Corollary 4.8. *Let S be a prescheme, H an S -group prescheme affine, flat and of finite presentation over S , with fibers of multiplicative type.*

For H to be of multiplicative type, it is necessary and sufficient that it satisfy one of the following two conditions (equivalent under the preceding conditions):

- a) *The type of H_s (cf. IX 1.4) is a locally constant function of $s \in S$.*
- b) *H is commutative, and the ${}_n H$ ($n > 0$) are finite over S .*

⁽³¹⁾**N.D.E.** The original has been expanded in the preceding and following passages.

Moreover, these conditions a), b) are implied by the following:

c) The fibers of H are connected.

In particular, one finds:

Corollary 4.9. *Let S be a prescheme, H an S -group prescheme flat and of finite presentation over S . Suppose in addition H affine over S with connected fibers.⁽³²⁾ If $s \in S$ is such that H_s is a torus, there exists an open neighborhood U of s such that $H|_U$ is a torus.*

In particular, if all the fibers of H are tori, H is a torus.

5. Scheme of homomorphisms from one group of multiplicative type into another. Twisted constant groups and groups of multiplicative type

Definition 5.1. a) Let S be a prescheme, R a group prescheme over S . One says that R is a *twisted constant group over S* if it is locally isomorphic, in the sense of the faithfully flat quasi-compact topology, to a constant group scheme, i.e. of the form M_S with M an ordinary group.

b) One says that the twisted constant group R over S is *quasi-isotrivial*, resp. *isotrivial*, resp. *locally isotrivial*, resp. *locally trivial*, resp. *trivial*, if in the above definition one can replace the faithfully flat quasi-compact topology by the étale topology, resp. the global finite étale topology, resp. the finite étale topology, resp. the Zariski topology, resp. the coarsest (or “chaotic”) topology, cf. IX 1.1 and 1.2, and IV 6.6.

Saying that R is quasi-isotrivial (resp. isotrivial) therefore means that there exists a surjective étale (resp. and finite) morphism $S' \rightarrow S$ such that $R' = R \times_S S'$ is a constant group over S ; saying that it is trivial means that R is a constant group. c) One defines as in VIII 1.4 the *type* of a twisted constant group R over S at an $s \in S$; it is an isomorphism class of ordinary groups which, for variable s , is a locally constant function of s , and hence constant if S is connected. One will also say that R is “of type M ” if all the fibers of R are of type M .

One should take care that R is quasi-compact over S only if it is finite over S , i.e. if its type at every $s \in S$ is a finite group.⁽³³⁾

d) The case of greatest interest to us is that where R is commutative. We shall then say that R is “finitely generated” if its type at every point $s \in S$ is given by a $\mathbf{Z}$ -module of finite type, a notion which must not be confused with the schematic notion “ R of finite type over S ” (cf. above).

Remark 5.2. We shall also have to consider S -preschemes X which are locally isomorphic (for the faithfully flat quasi-compact topology) to constant preschemes, independently of any group structure. We shall then say that X is a *twisted constant bundle over S* , and extend to these preschemes the terminology introduced in 5.1. Of course, one should take care that when X is endowed with an S -group structure, the meaning of the expressions “twisted constant”, “isotrivial”, etc. changes according to whether one takes account of the group structure over S . The same is true if one considers on X any other kind of algebraic structure (for example that of a Galois principal bundle, which will be considered in the next no.).

Proposition 5.3. *Let R be a commutative twisted constant group over S .*

(i) *The functor $H = D_S(R)$ (cf. VIII 1) is representable and is a group of multiplicative type over S .*

(ii) *For every $s \in S$, the type of R at s equals that of H at s .*

⁽³²⁾**N.D.E.** This result is generalized in 8.1: in fact it suffices to suppose the fibers of H affine and connected.

⁽³³⁾**N.D.E.** See lemma 5.12 below.

(iii) For R to be quasi-isotrivial (resp. isotrivial, resp. trivial, resp. locally isotrivial, resp. locally trivial), it is necessary and sufficient that H be so.

Remark 5.3.1. ⁽³⁴⁾ One may be concerned to see the term “type” used in 5.3 in two different senses according to whether R or H is involved; fortunately, when an S -group prescheme G is both a twisted constant group and of multiplicative type, its type in either sense is the same, thanks to the fact that an ordinary finite commutative group is isomorphic to its dual!

PROOF. As covering families for the faithfully flat quasi-compact topology are families of effective descent for the fibered category of affine group preschemes over variable base preschemes (SGA 1, VIII 2.1), one sees that H is representable (and affine over S), since it is so “locally”⁽³⁵⁾ (for it is so when R is constant, and then H is a diagonalizable group).

The fact that H is of multiplicative type is then evident by definition, as is the fact that the type of R and H at $s \in S$ is the same. Finally, since $H_{S'} = D_{S'}(R_{S'})$, the last assertion is reduced to the “trivial” case, i.e. to checking that R is trivial if and only if H is so, which follows immediately from biduality theorem VIII 1.2. $\square$

To finish specifying the correspondence between twisted constant groups and groups of multiplicative type, one must start with a group of multiplicative type H and study $R = D_S(H)$. If the latter is representable, it is evidently a twisted constant group, and one shall have $H \simeq D_S(R)$. In other words:

Scholium 5.4.0. *The functor $R \mapsto D_S(R)$ is an anti-equivalence⁽³⁶⁾ between the category of twisted constant groups over S and that of groups of multiplicative type H over S such that $D_S(H)$ is representable.*

I do not know whether this condition on H is always satisfied; we shall nevertheless see that it is when H is quasi-isotrivial, in particular when H is of finite type.

Lemma 5.4. *Let $S' \rightarrow S$ be a faithfully flat morphism locally of finite presentation, X' an S' -prescheme separated, locally of finite presentation and locally quasi-finite over S' .*

Then every descent datum on X' relative to $S' \rightarrow S$ is effective, i.e. there exists an S -prescheme X and an S' -isomorphism $X \times_S S' \simeq X'$ compatible with the descent datum.

One has already noted that the hypothesis on $S' \rightarrow S$ implies that it is a covering morphism for the faithfully flat quasi-compact topology (IV 6.3.1 (iv)), a fortiori a morphism of effective descent.

When $X' \rightarrow S'$ is quasi-compact, hence of finite presentation and quasi-finite, it is a quasi-affine morphism (cf. SGA 1, VIII 6.2⁽³⁷⁾), and effectiveness follows in this case from SGA 1, VIII 7.9. In the general case one reduces immediately to the case where S, S' are affine. One covers X' by affine open subsets U'_i ; let V'_i be the saturation of U'_i for the equivalence relation in X' defined by the descent datum, i.e. $V'_i = q_2(q_1^{-1}(U'_i))$, where q_1, q_2 are the two projections of $X'_1 = X' \times_{S'} (S'', p_1)$ onto X' ($q_1 = \text{pr}_1$, and q_2 is deduced from the first projection of $X'_2 = X' \times_{S'} (S'', p_2)$ by the given descent isomorphism $X'_1 \simeq X'_2$). Since $S' \rightarrow S$ is faithfully flat quasi-compact locally of finite presentation, so is $p_1 : S'' = S' \times_S S' \rightarrow S'$, hence so are q_1, q_2 , which are consequently open morphisms (SGA 1 IV 6.6⁽³⁸⁾). Consequently V'_i is an open and quasi-compact subset of X' . By what has already been seen, the descent data induced on the V'_i are effective, whence it follows that the datum on X' is also so (SGA 1, VIII 7.2).

⁽³⁴⁾**N.D.E.** This remark, which appeared after the proof in the original, has been moved here.

⁽³⁵⁾**N.D.E.** Cf. VIII §1.7.

⁽³⁶⁾**N.D.E.** “Equivalence” has been corrected to “anti-equivalence”.

⁽³⁷⁾**N.D.E.** (When S' is locally noetherian, and EGA IV₃, 8.11.2 in general.)

⁽³⁸⁾**N.D.E.** (When S' is locally noetherian, and EGA IV₂, 2.4.6 in general.)

Corollary 5.5. *A faithfully flat morphism of finite presentation $S' \rightarrow S$ is a morphism of effective descent for the fibered category of twisted constant groups (over variable base preschemes).*

Indeed, this amounts to asserting the effectiveness of a descent datum under the conditions of 5.4 when X' is a constant S' -prescheme.

Theorem 5.6. *Let S be a prescheme, G, H two S -group preschemes of quasi-isotrivial multiplicative type, with $G^{(39)}$ of finite type.*

Then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is representable by, and is, a quasi-isotrivial twisted constant group over S ; ⁽⁴⁰⁾ for every $s \in S$, if the type of G (resp. H) at s is M (resp. N), that of $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$.

One proceeds as in 4.2, using the fact that the assertion is established (VIII 1.5) when G, H are trivial. The necessary effectiveness criterion is supplied by 5.5 (in the case of a surjective étale morphism $S' \rightarrow S$).

In particular, putting $G = \mathbf{G}_{m,S}$, one finds:

Corollary 5.7. (i) *Let H be an S -group of quasi-isotrivial multiplicative type; then the S -group $D_S(H)$ is representable and is a quasi-isotrivial twisted constant group over S .*

(ii) *The functors $R \mapsto D_S(R)$ and $H \mapsto D_S(H)$ are anti-equivalences, quasi-inverse to one another, between the category of quasi-isotrivial twisted constant S -groups and that of S -groups of quasi-isotrivial multiplicative type.*

(iii) *These functors induce anti-equivalences between the subcategories consisting of isotrivial, resp. locally isotrivial, resp. locally trivial, resp. trivial groups.*

The last assertion is mentioned only for the record, being already contained in 5.3.

Moreover, since every S -group of multiplicative type and of finite type is quasi-isotrivial by 4.5, one deduces from 5.6:

Corollary 5.8. *Let S be a prescheme, G, H two S -group preschemes of multiplicative type and of finite type; then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$ is representable and is a finitely generated quasi-isotrivial twisted constant S -group over S .*

Note also that in 5.3, R is finitely generated if and only if $H = D_S(R)$ is of finite type (IX 2.1 b)). By 4.5, H is then quasi-isotrivial, so R is quasi-isotrivial. One thus finds:

Corollary 5.9. *The functors of 5.7 induce anti-equivalences quasi-inverse to one another between the category of S -groups H of multiplicative type of finite type and that of finitely generated twisted constant S -groups R ; moreover, every such group R is quasi-isotrivial.*

Corollary 5.10. *Let H, G be two S -group preschemes of multiplicative type and of finite type.*

(i) *Then $\underline{\mathrm{Isom}}_{S\text{-gr.}}(H, G)$ is representable by an open and closed subprescheme of $\underline{\mathrm{Hom}}_{S\text{-gr.}}(H, G)$, and is a twisted constant S -prescheme.*

(ii) *In particular, $\underline{\mathrm{Aut}}_{S\text{-gr.}}(H)$ is representable and is a twisted constant S -group (in general noncommutative).*

This follows from 5.8 and VIII 1.6. ⁽⁴¹⁾

Recollection 5.11.0. ⁽⁴²⁾ Recall that if X is a locally noetherian prescheme, its connected components (which are always closed) are open. Indeed, let C be a connected component

⁽³⁹⁾**N.D.E.** H has been corrected to G .

⁽⁴⁰⁾**N.D.E.** The following has been added.

⁽⁴¹⁾**N.D.E.** Correct this reference by treating the case of the functor $\underline{\mathrm{Isom}}_{S\text{-gr.}}(H, G)$ in an addition 1.5.1 in VIII...

⁽⁴²⁾**N.D.E.** This recollection has been added; it is used several times in what follows. (See also EGA I, 6.1.9; note, however, that the proof in loc. cit. seems unnecessarily complicated.)

of X , $x \in C$ and U a noetherian open neighborhood of x ; then U has only finitely many connected components, so the connected component U_x of x in U is open in U , hence in X ; as $U_x \subseteq C$, this shows that C is open in X .

Proposition 5.11. *Let S be a prescheme, R a commutative twisted constant group over S , $H = D_S(R)$ the group of multiplicative type it defines. Consider the following conditions:*

- (i) H is isotrivial (i.e. R is isotrivial).
 - (ii) R is a union of subpreschemes R_i which are both open and closed and quasi-compact over S (and then necessarily finite over S).
 - (iii) The connected components of R are finite over S .
- a) Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii).⁽⁴³⁾
 b) One has (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) if S is locally noetherian.
 c) Finally, (i) $\Leftrightarrow$ (ii) if R is finitely generated (i.e. if H is of finite type over S), at least if S is quasi-compact or its connected components are open.

First decomposing S as a sum prescheme of preschemes S_i on each of which H has constant type (cf. IX 1.4.1), one is reduced to the case where H , and hence R , has constant type M . We shall need the

Lemma 5.12. *Let S be a prescheme, R a twisted constant prescheme over S . Then every closed subprescheme Z of R which is quasi-compact over S is finite over S .*

Indeed, one is reduced to the case where R is constant, hence of the form I_S , where I is a set, hence the increasing filtering union of the J_S , where J ranges over the finite subsets of I . One may suppose in addition S affine; then Z is quasi-compact, hence contained in one of the J_S . Since J_S is finite over S , so is Z .

Lemma 5.12 already establishes the assertion in parentheses in 5.11 (ii).⁽⁴⁴⁾ The implication (ii) $\Rightarrow$ (iii) is then clear, since the connected components of R are closed; by 5.11.0 they are open and closed if S is locally noetherian (R being étale, hence locally of finite type over S), whence (iii) $\Rightarrow$ (ii) in this case.

Let us prove (i) $\Rightarrow$ (ii). For this, let $S' \rightarrow S$ be a finite surjective étale morphism splitting H , and hence R (cf. 5.3), so that R' is isomorphic to $M_{S'} = \coprod_{m \in M} R'_m$, where the R'_m are disjoint open subsets of R' , S' -isomorphic to S' . Let $g : R' \rightarrow R$ be the projection, which is a finite surjective étale morphism, hence an open and closed morphism; then the $R_m = g(R'_m)$ are subsets both open and closed in R , and evidently quasi-compact over S since the R'_m are so.

Finally, suppose H of finite type over S , and let us prove (ii) $\Rightarrow$ (i). The case where the connected components of S are open reduces immediately to the case where S is connected, so one may suppose S quasi-compact or connected. Since M is finitely generated, one may write $M = \mathbf{Z}^r \times N$, where r is an integer ≥ 0 and N a finite abelian group. Let $G = D_S(M)$; consider the preschemes

$$P = \underline{\text{Isom}}_{S\text{-gr.}}(H, G) \subset \underline{\text{Hom}}_{S\text{-gr.}}(H, G) = Q,$$

cf. 5.8 and 5.10. One has isomorphisms

$$\begin{aligned} Q &\simeq \underline{\text{Hom}}_{S\text{-gr.}}(M_S, R) \\ &\simeq \underline{\text{Hom}}_{S\text{-gr.}}(\mathbf{Z}_S^r, R) \times \underline{\text{Hom}}_{S\text{-gr.}}(N_S, R) \simeq R^r \times E, \end{aligned}$$

where $E = \underline{\text{Hom}}_{S\text{-gr.}}(N_S, R)$ is finite over S .⁽⁴⁵⁾ It follows that Q is a union of subpreschemes Q_i both open and closed and finite over S . Thus P is the union of the subpreschemes $P_i = P \cap Q_i$, both open and closed and finite over S . Since they are étale over S , their images in S are subsets S_i both open and closed, and they cover S . If S is connected or

⁽⁴³⁾**N.D.E.** Compare with examples 1.6. . .

⁽⁴⁴⁾**N.D.E.** The original has been expanded in what follows.

⁽⁴⁵⁾**N.D.E.** For it is a twisted constant group of type $\text{End}_{\text{gr.}}(N)$ (cf. 5.6 and 5.8).

quasi-compact, there therefore exists a finite set of indices i such that the S_i cover S ; let S' be the union of the corresponding P_i . Then $S' \rightarrow S$ is finite surjective étale, and putting $P' = P \times_S S' = \underline{\text{Isom}}_{S' \text{-gr.}}(H', G')$, one sees that P' has a section over S' , so there exists an isomorphism of S' -groups

$$H' = H \times_S S' \xrightarrow{\sim} G' = G \times_S S' = D_{S'}(M),$$

which proves that S' splits H . This completes the proof of 5.11.⁽⁴⁶⁾

Remark 5.11 bis. Let us note moreover that when H is of finite type over S and of constant type one may give the following isotriviality criterion (where it is no longer necessary to impose any restriction on S): $H = D_S(R)$ is isotrivial if and only if R is a union of a sequence of subsets both open and closed and finite over S .⁽⁴⁷⁾

Lemma 5.13. *Let S be a locally noetherian connected prescheme, P a quasi-isotrivial twisted constant S -prescheme, Z an open and closed subset of P such that there exists an $s \in S$ with Z_s finite. Then Z is finite over S .*

Let us not initially suppose S connected: let U be the set of $s \in S$ such that Z_s is finite; we shall prove that U is both open and closed, and that $Z|_U$ is finite over U . This is evidently a statement essentially equivalent to 5.13 but with the advantage of being “of local nature” on S for the étale topology (say), which reduces us to the case where P is constant, i.e. of the form I_S , where I is a set. (N.B. The locally noetherian hypothesis is not lost by an étale base change; this is where one uses the quasi-isotriviality of P over S .)

One may suppose in addition S connected, since its connected components are open (S being locally noetherian). But then one necessarily has $Z = J_S$, where J is a subset of I , and hence $U = \emptyset$ or $U = S$ according as J is infinite or finite, which gives the desired conclusion.

Recollections 5.14.0.⁽⁴⁸⁾ Let S be a locally noetherian prescheme, and let $\tilde{S}$ be the normalization of S_{red} (cf. EGA II, 6.3.8). Recall that S is said to be *geometrically unibranch* (cf. EGA 0_{IV}, §23.2 and IV₂, §6.15) if the canonical morphism $\tilde{S} \rightarrow S$ is radicial (and hence a universal homeomorphism); in particular, the connected components of S are irreducible.

Suppose then S connected, hence irreducible; let η be its generic point and let $f : P \rightarrow S$ be a flat and locally quasi-finite morphism. Let P_i be the irreducible components of P , and ξ_i the generic point of P_i . Since P is flat over S , each ξ_i lies over η (cf. EGA IV₂, 2.3.4), and therefore $(P_i)_\eta = P_i \cap P_\eta$ is the closure of ξ_i in P_η . Since the fiber P_η is discrete, one therefore has $(P_i)_\eta = \{\xi_i\}$. This applies in particular when f is étale; in this case P is also locally noetherian and geometrically unibranch (cf. EGA IV₄, 17.5.7), so its irreducible components are its connected components and are open (and closed).

Corollary 5.14. *Let S be a locally noetherian geometrically unibranch prescheme, P a quasi-isotrivial twisted constant S -prescheme. Then the connected components of P are finite over S .*

⁽⁴⁶⁾**N.D.E.** Modulo verification that (ii) $\Rightarrow$ (i) when S is locally noetherian and R is not finitely generated...

⁽⁴⁷⁾**N.D.E.** Expand this point: M being a $\mathbf{Z}$ -module of finite type, it is countable, and the proof of 5.11 (i) $\Rightarrow$ (ii) shows that R is a union of a countable family of open and closed subsets finite over S . One should see the converse...

⁽⁴⁸⁾**N.D.E.** The original treated in 5.14 the case where S is locally noetherian and normal, and pointed out in Remark 5.15 that the argument applies more generally if one merely supposes S geometrically unibranch instead of normal. The statement of 5.14 (and also 5.16) has been modified accordingly, and these “recollections” taken from EGA IV₄, 18.10.6 and 18.10.7 have been added, showing that the proof of 5.14 applies verbatim to the geometrically unibranch case.

One may evidently suppose S connected, hence irreducible; let η be its generic point. By the preceding, each connected component P_i of P is open and closed and meets the fiber P_η in a single point. Thus 5.13 applies and shows that each P_i is finite over S .

Theorem 5.16. ⁽⁴⁹⁾ *Let S be a locally noetherian geometrically unibranch prescheme. Then every S -group of multiplicative type and of finite type is isotrivial.*

Indeed, one may suppose S connected, hence H of constant type M . It suffices to apply 5.14 to $P = \underline{\text{Isom}}_{S\text{-gr.}}(H, G)$, where $G = D_S(M)$, then to argue as in the proof of 5.11 (ii) $\Rightarrow$ (i). One may also apply 5.14 to $P = R = D_S(H)$ (cf. 5.9), then use 5.11.

6. Infinite Galois principal coverings and the enlarged fundamental group

(The results of the present no. and the next will no longer be used in the rest of this Seminar.)

Let S be a prescheme; we propose to determine the principal homogeneous bundles P over S with structural group of the form G_S , the constant S -group defined by an ordinary group G (not necessarily finite), which one shall also call *Galois principal bundles over S with group G* . We take “principal bundle” in the sense of the faithfully flat and quasi-compact topology (cf. Exp. IV, Def. 5.1.5), but one will note that for such a P , the structural morphism $P \rightarrow S$ is necessarily étale and surjective, hence covering for the étale topology, so P is also locally trivial for the étale topology (cf. IV, Prop. 5.1.6).⁽⁵⁰⁾

We shall suppose that S is a sum of connected preschemes, i.e. that its connected components are open, which reduces us immediately to the case where S is connected. We shall then choose a “geometric point” ξ of S , i.e. an S -scheme ξ which is the spectrum of an algebraically closed field $\Omega = \kappa(\xi)$. Then for every Galois principal bundle P over S with group G , P_ξ is a Galois principal bundle over the algebraically closed field $\kappa(\xi)$, and hence trivial. We shall therefore specify the initial problem by proposing to determine the category of Galois principal bundles over S pointed over ξ , i.e. endowed with an S -homomorphism $\xi \rightarrow P$, i.e. with a trivialization of P_ξ . For fixed G , the set of classes of such bundles up to an isomorphism respecting the base point will be denoted by $\pi^1(S, \xi; G)$. Then the set $\pi^1(S; G)$ of isomorphism classes of Galois principal bundles over S with group G (without specified base point) is isomorphic to the set of orbits of G in $\pi^1(S, \xi; G)$ (taking account of the natural operations of G on this set, corresponding to translation by G of the marked point in a pointed Galois principal bundle P):

$$\pi^1(S; G) = \pi^1(S, \xi; G)/G.$$

For every morphism $S' \rightarrow S$ which is a morphism of universal effective descent for the fibered category of twisted constant preschemes over a variable base (for example $S' \rightarrow S$ faithfully flat and locally of finite presentation, cf. 5.4; for other examples cf. SGA 1 IX), we propose to determine the subsets of the preceding sets, denoted by $\pi^1(S'/S, \xi; G)$ and $\pi^1(S'/S; G)$, consisting of Galois principal bundles over S which become trivial over S' (or, as one says, are “split” by S'). One will in fact determine the category of Galois principal bundles P over S split by S' . Of course, one shall then have

$$\pi^1(S, \xi; G) = \varinjlim_{S'} \pi^1(S'/S, \xi; G),$$

where on the right-hand side S' ranges over a cofinal system in the set of S'/S covering for the étale topology (for example, when S is quasi-compact, the set of S' over S which are quasi-compact with étale and surjective structural morphism). Similarly, the category of

⁽⁴⁹⁾**N.D.E.** Remark 5.15, rendered obsolete by addition 5.14.0 (cf. the preceding N.D.E.), has been deleted, and in 5.16 “normal” has been replaced by “geometrically unibranch”.

⁽⁵⁰⁾**N.D.E.** Note that P is supposed to be a prescheme — in the a priori more general case of an (fpqc) sheaf P which is a G_S -torsor, is P necessarily representable?

Galois principal bundles over S will be the inductive limit of the subcategories defined by the S' (consisting of bundles split over S').

Thanks to the hypothesis on S'/S , the category of Galois principal bundles over S split by S' is equivalent to the category of trivial Galois principal bundles over S' (hence of the form $G_{S'}$, where G acts by right translations), endowed with a descent datum relative to $S' \rightarrow S$. Giving a base point on a Galois principal bundle P over S split by S' amounts, in terms of the corresponding trivial bundle P' over S' and its descent datum, to giving a trivialization of $P' \times_{S'} S'_\xi$ compatible with the induced descent datum relative to $S'_\xi \rightarrow \xi$ (N.B. One has put $S'_\xi = S' \times_S \xi$), i.e. a section σ of P'_ξ over S'_ξ compatible with the descent datum. It is then useful, for any S -prescheme S' (for which one no longer supposes that $S' \rightarrow S$ is a morphism of universal effective descent for the fibered category of twisted constant bundles...), to define $\pi^1(S'/S; G)$ and $\pi^1(S'/S, \xi; G)$ as the set of isomorphism classes of the structures with descent data just specified. One then obtains for these functors in G a very simple simplicial description, in terms of the fibered square and cube S'', S''' of S' over S , which we shall sketch below (cf. 6.3).

The important conclusion to retain will be the following:

Proposition 6.1. *Suppose that the connected components of S' and S'' are open, and, for example, that the quotient set of $\pi_0(S')$ by the equivalence relation induced by the two projections $\pi_0(S'') \rightarrow \pi_0(S')$ consists of one point.⁽⁵¹⁾*

(i) *The functor $G \mapsto \pi^1(S'/S, \xi; G)$ from the category of groups to the category of sets is representable by a group, denoted by $\pi_1(S'/S, \xi)$ and called the fundamental group of S at ξ relative to $S' \rightarrow S$. One therefore has a functorial bijection:*

$$\pi^1(S'/S, \xi; G) \simeq \text{Hom}_{\text{gr.}}(\pi_1(S'/S, \xi), G).$$

(ii) *This group has a set of generators in bijection with $\pi_0(S'')$, and is described in terms of these generators by relations in bijection with the elements of $\pi_0(S''')$.⁽⁵²⁾ In particular, $\pi^1(S'/S, \xi)$ is finitely generated (resp. of finite presentation) if $\pi_0(S'')$ (resp. as well as $\pi_0(S''')$) is finite. (iii) *The category of Galois principal bundles over S split by S' , with base point over ξ , is equivalent to the category of ordinary groups G endowed with a homomorphism from $\pi_1(S'/S, \xi)$ to G .**

The proof is given below, cf. ...

6.2. — When S is a connected locally noetherian prescheme, which implies that every étale prescheme S' over S is locally noetherian and hence has open connected components, one concludes from the preceding⁽⁵³⁾ that the functor $G \mapsto \pi^1(S, \xi; G)$ from the category of ordinary groups to the category of sets is strictly pro-representable (cf. Séminaire Bourbaki, February 1960, no. 195, §§ A.2 and A.3), i.e. there exists a projective system

$$\Pi = \Pi_1(S; \xi) = (\pi_i)_{i \in I}$$

⁽⁵¹⁾**N.D.E.** The hypothesis that the connected components of S' are open has been added, as has the second hypothesis. The latter means that the simplicial set $\mathbf{K}_\bullet = \pi_0(S_\bullet)$ defined in 6.3 is connected; in fact a weaker hypothesis suffices, namely that the cone $\bar{\mathbf{K}}_\bullet$ of a certain morphism of simplicial sets $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$ be connected (cf. loc. cit.).

⁽⁵²⁾**N.D.E.** The superfluous hypothesis that the connected components of S''' are open has been deleted. On the other hand, the description given below (cf. ...) gives as natural set of generators the set $\pi_0(S'') \coprod \pi_0(S'_\xi)$; one then reduces to $\pi_0(S'')$ by means of the relations between these generators coming from the 2-cells. See, for example, [Kan58], §19 for a finer description.

⁽⁵³⁾**N.D.E.** One could expand this deduction; I is in bijection with a cofinal set of morphisms $S' \rightarrow S$ covering for the étale topology...

of ordinary groups over an increasing filtering index set I which is “strict” (i.e. with surjective transition morphisms $\pi_j \rightarrow \pi_i$), and an isomorphism of functors in G

$$\pi^1(S, \xi; G) \simeq \varinjlim_i \operatorname{Hom}_{\operatorname{gr.}}(\pi_i, G).$$

The right-hand side is also simply denoted by $\operatorname{Hom}_{\operatorname{pro-gr.}}(\Pi, G)$ (cf. loc. cit.).

In the case where the projective limit $\pi = \varprojlim \pi_i$ is “sufficiently large”, more precisely when the canonical homomorphisms $\Pi \rightarrow \Pi_i$ are surjective, there is occasion to endow Π with the projective limit topology of the discrete topologies of the Π_i , and the preceding isomorphism is also written:

$$\pi^1(S, \xi; G) \simeq \operatorname{Hom}_{\operatorname{gr. top.}}(\pi, G),$$

where the right-hand side denotes the set of homomorphisms of topological groups, it being understood that G is endowed with the discrete topology. The hypothesis just formulated on the projective system Π is satisfied, as is well known, when the π_i are finite groups (cf. [BEns], III §7.4, Th. 1). This latter condition evidently also means that every Galois principal bundle over S is isotrivial, i.e. split by a finite surjective étale morphism. This is the case when S is geometrically unibranch (for example normal), as follows immediately from 5.14.⁽⁵⁴⁾ In the case where the π_i are finite, the group π also coincides with the fundamental group $\pi_1(S, \xi)$ introduced in SGA 1, V.

Thus, in the favorable case ($\pi \rightarrow \pi_i$ surjective), one could call π the *enlarged fundamental group* of S at ξ . Outside this favorable case, π itself is of little interest, and the role of the usual fundamental group is played by the projective system Π itself, which one shall call the *enlarged fundamental pro-group* of S at ξ . (Any terminological suggestion better than “enlarged” is welcome!⁽⁵⁵⁾) One will note that knowledge of this pro-group is more precise than that of the usual fundamental group $\pi_1(S, \xi)$ of SGA 1 V; more precisely, the latter is the projective limit of the projective system consisting of the finite quotients of the π_i .

6.3. — Let us indicate briefly the “calculation” of $\pi_1(S'/S, \xi)$. Let S_i be the $(i+1)$ -st fibered power of S' over S (i.e. $S_0 = S'$, $S_1 = S''$, etc.). There are evident simplicial operations between the S_i , making $(S_i)_{i \in \mathbb{N}}$ a simplicial object of $(\operatorname{Sch})/S$.

Transforming this simplicial object by the functor “set of connected components”

$$\pi_0 : (\operatorname{Sch})/S \rightarrow (\operatorname{Ens}),$$

one finds a simplicial set $\mathbf{K}_\bullet = (K_i)_{i \in \mathbb{N}}$, with $K_i = \pi_0(S_i)$.

Similarly, the $S_{i\xi}$ (= the $(i+1)$ -st fibered power of S'_ξ over ξ) form a simplicial object of $(\operatorname{Sch})/\xi$, hence of $(\operatorname{Sch})/S$, moreover endowed with a natural homomorphism of simplicial objects into $(S_i)_{i \in \mathbb{N}}$, whence a simplicial set $\mathbf{k}_\bullet$ (with $k_i = \pi_0(S_{i\xi})$) and a canonical homomorphism

$$\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet.$$

We may form a new simplicial set by taking the cone of this morphism (cf. 9.5.1):

$$\tilde{\mathbf{K}}_\bullet = \operatorname{Cone}(\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet).$$

⁽⁵⁶⁾ In this way one obtains a “pointed simplicial set” $\tilde{\mathbf{K}}_\bullet$ (i.e. a simplicial set endowed with a homomorphism $\tilde{\xi} : \mathbf{e}_\bullet \rightarrow \tilde{\mathbf{K}}_\bullet$, where $\mathbf{e}_\bullet$ is the final simplicial set). We may construct its well-known combinatorial invariants $\pi_0(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ and $\pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$, whose construction moreover involves only the components of degree ≤ 1 resp. of degree ≤ 2 . These invariants are defined without restriction on S or S' . One verifies without difficulty, when the connected

⁽⁵⁴⁾N.D.E. 5.13 has been corrected to 5.14.

⁽⁵⁵⁾N.D.E. Some authors speak of the “true” fundamental group.

⁽⁵⁶⁾N.D.E. The original has been corrected: it considers $\tilde{\mathbf{K}}_\bullet = \mathbf{K}_\bullet/\mathbf{k}_\bullet$, whereas on the one hand $\mathbf{k}_\bullet$ is already contractible (cf. 9.7.1), and on the other hand the morphism $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$ is not in general injective. It is an epimorphism if S'/S is finite étale (cf. 9.7.3).

components of S_0, S_1 are open and $\tilde{\mathbf{K}}_\bullet$ is connected,⁽⁵⁷⁾ that $\pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ represents the functor $G \mapsto \pi^1(S'/S, \xi; G)$, i.e. that one has:

$$\pi_1(S'/S; \xi) \simeq \pi_1(\tilde{\mathbf{K}}_\bullet, \tilde{\xi}).$$

Let us also point out that when the morphism $S' \rightarrow S$ is “universally submersive” (cf. SGA 1, IV 2.1), and the connected components of S' are open,⁽⁵⁸⁾ the simplicial set $\mathbf{K}_\bullet$, and hence also $\tilde{\mathbf{K}}_\bullet$, is connected.

Examples 6.4. It remains to give examples of enlarged fundamental groups. Return to examples 1.6, i.e. let k be an algebraically closed field, C_1 a complete rational curve over k having exactly one singular point, this point being an ordinary double point, and C_2 a curve which is a union of two irreducible components isomorphic to $\mathbb{P}_k^1$ and meeting at exactly two points, which are ordinary double points of C_2 . In either case the enlarged fundamental group of the curve is isomorphic to $\mathbf{Z}$.⁽⁵⁹⁾

In general, there would be occasion to reconsider (simplifying and correcting them) the results of SGA 1 IX 5 in the setting of the enlarged fundamental group. The examples of loc. cit., 5.5 would give as many examples of enlarged fundamental pro-groups which are not profinite. Thus, if instead of an ordinary double point one took in the first example a double point with n distinct branches,⁽⁶⁰⁾ one would find as enlarged fundamental group the free (discrete!) group on $n - 1$ generators.

7. Classification of twisted constant preschemes and groups of multiplicative type of finite type in terms of the enlarged fundamental group

7.0. — Let S be a prescheme, which we again suppose locally noetherian, to ensure that S and certain preschemes over S which we shall consider (in particular those étale over S , more generally those locally of finite type over S) are locally connected.

Proposition 7.0.1. *Every twisted constant prescheme X over S which is locally trivial for the (fppf) topology (i.e. split by a faithfully flat morphism locally of finite presentation $S' \rightarrow S$) is quasi-isotrivial (i.e. one may even choose $S' \rightarrow S$ étale surjective).*

Indeed, one may suppose S connected, hence X of type I , where I is a fixed set. Thus $X' = X \times_S S'$ is isomorphic to $I_{S'}$, so $I_{S'}$ is endowed with a descent datum relative to $S' \rightarrow S$, i.e. one has an isomorphism $I_{S''} \xrightarrow{\sim} I_{S''}$ satisfying the usual transitivity condition. Now $S'' = S' \times_S S'$ is locally noetherian, hence locally connected, whence it follows that the automorphisms of $I_{S''}$ correspond to the sections of $G_{S''}$, where $G = \text{Aut}(I)$ is the permutation group of I .

In this way one obtains a descent datum on $G_{S'}$ (regarded as a trivial Galois principal bundle) relative to $S' \rightarrow S$. By 5.4 this descent datum is effective, whence a Galois principal bundle P over S with group G . By construction it represents the functor $\underline{\text{Isom}}_S(I_S, X)$ in

⁽⁵⁷⁾**N.D.E.** The hypothesis that $\tilde{\mathbf{K}}_\bullet$ is connected has been added; for the proof, see the addenda below (section 9).

⁽⁵⁸⁾**N.D.E.** What follows has been modified, taking account of the correction made above, cf. N.D.E. (53). (The original was: “ $\pi_0(\tilde{\mathbf{K}}_\bullet, \tilde{\xi})$ is also canonically isomorphic to the set $\pi_0(S, \xi)$ of connected components of S , pointed by the connected component of ξ in S ”.)

⁽⁵⁹⁾**N.D.E.** One could expand this: first, in view of 5.14, the enlarged fundamental group of the projective line $\mathbb{P}_k^1$ is zero, i.e. $\mathbb{P}_k^1$ is “truly” simply connected. Then one should extend to the case of the enlarged fundamental group and of the category of principal bundles (not necessarily finite) Corollary 5.4 of SGA 1 IX and the discussion following it. (Let Γ_1, Γ_2 be two copies of $\mathbb{P}_k$, a_i, b_i two distinct points of Γ_i , C_2'' the disjoint union of Γ_1, Γ_2 , C_2' the curve obtained by identifying a_1 with a_2 ; then C_2 is obtained from C_2' by further identifying b_1 with b_2 . The discussion following loc. cit., extended to the enlarged case, then shows that the enlarged fundamental group of C_2' (resp. C_2) is zero (resp. $\mathbf{Z}$).)

⁽⁶⁰⁾**N.D.E.** This is an n -fold point with n distinct tangents, for example the curve $X^n - Y^n = X^{n+1}$.

the category of locally noetherian preschemes over S . Consequently, the surjective étale base change $P \rightarrow S$ splits X , so X is indeed quasi-isotrivial.

Remark 7.0.2. One should take care that even if S is the spectrum of a field, it is not true in general that every twisted constant bundle over S is quasi-isotrivial. It suffices, for example, to take for X the sum scheme of a sequence of schemes of the form $\mathrm{Spec}(k_i)$, where the k_i are separable extensions of k of strictly increasing degrees.

The proof given above shows at the same time that classification of quasi-isotrivial twisted constant bundles X over S of type I is equivalent to that of Galois principal bundles over S with group $G = \mathrm{Aut}(I)$. This is even an equivalence of categories.

It may be put in a more convenient form as in SGA 1 V. For this, suppose S connected and endowed with a geometric point ξ . Consequently the enlarged fundamental pro-group $\Pi = \Pi_1(S, \xi)$ is defined. On the other hand, for every quasi-isotrivial twisted constant bundle X over S , let $I = X(\xi)$ be its set-theoretic fiber at ξ . Thus X is of type I , and consequently associated, as just stated, with a Galois principal bundle $P = \underline{\mathrm{Isom}}_S(I_S, X)$ over S with group $G = \mathrm{Aut}(I)$.

By the definition of Π , one therefore obtains a canonical homomorphism from Π to G , i.e. from one of the Π_i to G . Since G is the permutation group of $I = X(\xi)$, this means that Π “acts continuously on $I = X(\xi)$ ”, it being understood by this that the π_i (large i) act on I compatibly with the transition morphisms.

We leave it to the reader to check that every S -morphism $X \rightarrow Y$ between quasi-isotrivial twisted constant bundles over S induces a map $X(\xi) \rightarrow Y(\xi)$ compatible with the operations of Π , and that the functor so obtained is an equivalence of categories:

Proposition 7.0.3. *Let S be a connected locally noetherian prescheme, ξ a geometric point of S , $\Pi = \Pi_1(S, \xi)$ the enlarged fundamental pro-group of S at ξ . Then the functor*

$$X \mapsto X(\xi)$$

is an equivalence between the category of quasi-isotrivial twisted constant bundles over S and the category of sets on which Π acts continuously.

This functor is compatible with the operations of finite sums and finite projective limits. It follows, for example, that quasi-isotrivial twisted constant groups (or rings, etc.) over S correspond to ordinary groups (resp. rings, etc.) on which the pro-group Π acts continuously. In particular:

Corollary 7.0.4. *The category of quasi-isotrivial twisted constant commutative groups over S is equivalent to the category of “ Π -modules”, i.e. of commutative groups M on which Π acts continuously.*

Using now 5.7, one concludes the:

Theorem 7.1. *Let S be a connected locally noetherian prescheme, ξ a geometric point of S , $\Pi = \Pi_1(S, \xi)$ the enlarged fundamental pro-group of S at ξ . Then the functor*

$$G \mapsto \mathrm{Hom}_{\kappa(\xi)\text{-gr.}}(G_\xi, \mathbf{G}_{m, \xi})$$

induces an anti-equivalence of the category of quasi-isotrivial groups of multiplicative type over S with the category of Π -modules.

Using 4.5 one concludes:

Corollary 7.2. *The preceding functor induces an anti-equivalence of the category of groups of multiplicative type and of finite type over S and the category of Π -modules of finite type over $\mathbf{Z}$.*

Example 7.3. Take again, for example, for S a complete rational curve over an algebraically closed field, having exactly one multiple point with $n+1$ distinct branches. By 6.4 the enlarged fundamental group $\Pi(S, \xi)$ is a free group on n generators. Thus by 7.2, classification of tori of relative dimension m over S is equivalent to classification of systems of n endomorphisms $A_1, \dots, A_n$ of the $\mathbf{Z}$ -module $\mathbf{Z}^m$, up to automorphism of $\mathbf{Z}^m$. Except for $n \leq 1$ or $m \leq 1$, explicit classification of such systems seems hopeless. One may at least define a host of nontrivial invariants for such a system, such as the characteristic polynomials of the A_i .⁽⁶¹⁾

Remark 7.4. If no hypothesis is made on S , it remains true that for a given ordinary commutative group M of finite type over $\mathbf{Z}$, the category of groups of multiplicative type of type M over S is anti-equivalent to the category of Galois principal bundles over S with group $G = \text{Aut}_{\text{gr.}}(M)$. This follows easily from 5.9 and 5.10.

Remark 7.5. The theory of the fundamental pro-group which we have sketched in the present two nos. will advantageously be written in the setting of general sites. In this form it also applies, for example, to ordinary topological spaces, and gives a satisfactory theory at least for a locally connected topological space (not necessarily locally simply connected). In this case too it seems that one cannot content oneself with defining a fundamental group and that a pro-group is needed. Finally, let us note that once one has the language of topologies and descent (which is really at the heart of these questions), the exposition sketched here is also technically simpler than that of SGA 1 V, and should therefore in principle replace it.

8. Appendix: Elimination of certain affineness hypotheses

Our object is to prove the following generalization of 4.9.

Theorem 8.1. *Let S be a prescheme, H an S -group prescheme flat and of finite presentation, with connected and affine fibers.⁽⁶²⁾ Let $s \in S$; suppose H_s is a torus. Then there exists an open neighborhood U of s such that $H|U$ is a torus.*

One concludes immediately:

Corollary 8.2. *Let H be an S -group prescheme flat and of finite presentation over S . For H to be a torus, it is necessary and sufficient that its fibers be tori.*

Remark 8.3. Even when S is the spectrum of a discrete valuation ring, one cannot in 8.1 drop the hypothesis that the fibers of H (here the generic fiber) are affine, for there are examples of smooth groups over S whose generic fiber is an elliptic curve and whose special fiber is $\mathbf{G}_m$.

PROOF OF 8.1. One may evidently suppose $S = \text{Spec}(A)$ affine, which reduces us by the standard procedure (cf. EGA IV₂, 8.8.2) to the case where S is in addition noetherian. We begin by proving 8.2 in this case.

By 4.9 one is reduced to proving that H is affine over S . One may therefore suppose⁽⁶³⁾ A noetherian local, and since the completion $\hat{A}$ is faithfully flat over A , one is reduced by descent to the case where A is a complete noetherian local ring. Let S' be the normalization of S_{red} ; one knows by Nagata that it is finite over S (EGA 0_{IV}, 23 1.5); moreover $S' \rightarrow S$ is surjective, so $H' = H \times_S S' \rightarrow H$ is finite and surjective; hence, to prove that H is affine, it suffices to show that H' is so (EGA II, 6.7.1).

⁽⁶⁴⁾ Replacing S by a connected component of S' reduces us to the case where A is a noetherian (semilocal) normal integral ring. Moreover, replacing A by its normalization in

⁽⁶¹⁾**N.D.E.** In particular, this shows that the tori of relative dimension 2 considered in 1.6 are nonisotrivial.

⁽⁶²⁾**N.D.E.** Note that the hypotheses imply that H is separated over S , by VI_B, Th. 5.3 or Cor. 5.5.

⁽⁶³⁾**N.D.E.** By EGA IV₂, 8.8.2 again.

⁽⁶⁴⁾**N.D.E.** The original has been expanded in what follows.

a finite separable extension of its field of fractions if necessary, one may suppose that the generic fiber H_η of H is diagonalizable, i.e. that one has an isomorphism

$$u_\eta : H_\eta \xrightarrow{\sim} T_\eta,$$

where $T = \mathbf{G}_{m,S}^r$. Now one has the

Lemma 8.4. *Let S be a locally noetherian normal irreducible prescheme with generic point η , H a group prescheme over S smooth with connected fibers, T a group prescheme of multiplicative type and of finite type over S , $u_\eta : H_\eta \rightarrow T_\eta$ a homomorphism of algebraic groups over $\kappa(\eta)$.*

Then u_η extends to a group homomorphism $u : H \rightarrow T$.

Replacing S by its normalization in a finite separable extension of its function field if necessary, one may suppose T diagonalizable (which is moreover the case in the application we have in mind). Then T is a closed subgroup of a group of the form $\mathbf{G}_{m,S}^r$, which reduces us to the case where $T = \mathbf{G}_{m,S}$.

Everything amounts to proving that u_η , regarded as a rational map from H to $T = \mathbf{G}_{m,S}$, is everywhere defined (for the morphism $u : H \rightarrow T$ extending it is then necessarily a group homomorphism). One may regard u_η as an invertible section f of the structure sheaf of H_η , and must show that it extends to an invertible section of the structure sheaf of H . Now H , being smooth over normal S , is normal (SGA 1, II 3.1), so it suffices to find a closed subset Z of H of codimension ≥ 2 such that f extends to an invertible section of the structure sheaf of $H - Z$. This reduces us immediately to the case where S is the spectrum of a discrete valuation ring A (by localizing at the codimension 1 points of S).

Let t be a uniformizer of A , t' the section of $\mathcal{O}_H$ it defines, so that the special fiber H_0 equals $\mathcal{V}(t')$. By hypothesis H_0 is smooth over the residue field k and connected. Then f is a rational function on H with neither zeros nor poles in $H - H_0$; since $H_0 = \mathcal{V}(t')$ is an irreducible divisor, there exists an integer $n \in \mathbf{Z}$ such that $t'^n f = t^n f$ has neither zeros nor poles, i.e. is an invertible section of $\mathcal{O}_H$. It therefore defines a morphism $v : H \rightarrow \mathbf{G}_{m,S}$, and since $v_\eta = t'^n u_\eta$ and u_η is a group homomorphism, v transforms the unit section of H into a section of $\mathbf{G}_{m,S}$ whose value at the generic point of S is t^n ; since it is a section of $\mathbf{G}_{m,S}$, t^n must be a unit, i.e. $n = 0$, so v extends u_η , which finishes proving 8.4.

Applying this lemma to the present case, one finds a group homomorphism

$$u : H \rightarrow T = \mathbf{G}_{m,S}^r$$

inducing an isomorphism on the generic fibers. Let us prove that u is an isomorphism.

Lemma 8.5. *For every integer $n > 0$ prime to the residue characteristic of A , ${}_n H = \text{Ker}(n \cdot \text{id}_H)$ is finite over S .*

If n is prime to the residue characteristic of A , it is prime to all the residue characteristics of the points of S . Thus $n \cdot \text{id}_H$ induces on every fiber of H an étale morphism, so $n \cdot \text{id}_H$ is étale (SGA 1, I 5.9), hence its kernel ${}_n H$ is étale over S . On the other hand, ${}_n H$ is separated over S since H is so.⁽⁶⁵⁾⁽⁶⁶⁾ Moreover, all its fibers have the same rank n^r , since the fibers of H are tori, all of the same dimension r (H being smooth over S). One concludes that ${}_n H$ is finite over S (SGA 1, I 10.9 or EGA IV₄, 18.2.9).

Thus by the preceding lemma, $u({}_n H)$ is a closed subset of T . Since on the generic fiber T_η this subset is identical to ${}_n(T_\eta)$, it follows that it contains the closure of this subset, namely ${}_n T$. Now for every $s \in S$, the ${}_n(T_s)$ are dense in T_s ; as $u_s(H_s)$ is a closed subset (VI_B 1.4.2) containing them, one sees that $u_s(H_s) = T_s$, hence u is surjective. Since a surjective

⁽⁶⁵⁾By Raynaud's theorem VI_B 5.3.

⁽⁶⁶⁾**N.D.E.** Expand this point, in connection with the modifications in VI_B §5.

homomorphism of tori of the same dimension over a field is flat,⁽⁶⁷⁾ it follows that u induces a flat morphism on every fiber, so u is flat (SGA 1 I 5.9). Consequently $K = \text{Ker}(u)$ is flat over S ,⁽⁶⁸⁾ hence equal to the closure of its generic fiber K_η . Now K_η is the unit group by construction, and since K is separated over S (since H is so), its unit section is closed, whence it follows that K is the unit group. Thus u is a monomorphism; as one has seen it to be faithfully flat, it is therefore an isomorphism (cf. SGA 1, I 5.1 or EGA IV₄, 17.9.1). This proves that H is a torus, and hence completes the proof of 8.2.⁽⁶⁹⁾

Remark 8.5.1. Instead of invoking 8.5, one may also invoke Zariski's "Main Theorem", which directly implies that u is an open immersion, hence an isomorphism.⁽⁷⁰⁾

To prove 8.1, one is reduced, thanks to the quasi-isotriviality theorem, to the case where S is local and s its closed point,⁽⁷¹⁾ and to proving that under the other hypotheses made H is then a torus. By 8.2 already proved, one is reduced to proving that the fibers of H are tori. One may suppose S the spectrum of a complete noetherian local ring A . By 3.3 there exists for every $n > 0$ a finite group of multiplicative type T_n over S and an isomorphism $(T_n)_s \simeq {}_n(H_s)$, where s is the closed point of S . Proceeding as in 3.1 and using the fact that T_n is finite over S ,⁽⁷²⁾ one sees that the preceding isomorphism comes from a homomorphism $u_n : T_n \rightarrow H$, moreover uniquely determined. (Pass to the limit over the artinian quotients of A .)

Moreover, by the uniqueness properties, $u_n : T_n \rightarrow H$ is deduced from $u_m : T_m \rightarrow H$ by restriction to ${}_n(T_m) \simeq T_n$ when m is a multiple of n . It follows from IX, 6.6 that the u_n are monomorphisms, so the T_n are subgroups of H , and for m a multiple of n one has $T_n = {}_n(T_m)$.

Thus for every $t \in S$, the $(T_n)_t$ are subgroups of H_t of type $(\mathbf{Z}/n\mathbf{Z})^r$ (where $r = \dim H_s = \dim H_t$), such that for m a multiple of n one has $(T_n)_t = {}_n(T_m)_t$. The fact that H_t is a torus now follows from the

Lemma 8.6. *Let H be a smooth affine algebraic group over an algebraically closed field k , $(T_n)_{n>0}$ a family of subgroups of multiplicative type of H such that for every integer $n > 0$, T_n is of type $(\mathbf{Z}/n\mathbf{Z})^r$, and for every multiple m of n one has $T_n = {}_n(T_m)$.*

Under these conditions, H contains a torus of dimension $\geq r$ containing the T_n , so if H is connected of dimension $\leq r$, H is a torus of dimension r .

This is an exercise on affine algebraic groups which we shall treat by references to *Bible*. We restrict to considering the T_n for n prime to the characteristic. Let K be the closure of the union of the T_n in H , endowed with the induced reduced structure; then standard arguments show that K is a commutative algebraic subgroup of H . By *Bible*, §4.5, th. 4, K is therefore isomorphic to a product $K_u \times K_s$, with K_u "unipotent" and K_s diagonalizable. Every diagonalizable subgroup of K is contained in K_s , so the T_n are subgroups of K_s , hence $K = K_s$. Write $K = D(M)$ with M an ordinary commutative group of finite type over $\mathbf{Z}$; then $T_n \subseteq K$ means that M admits a quotient group isomorphic to $(\mathbf{Z}/n\mathbf{Z})^r$. This being true for every integer n prime to the characteristic of k (it would suffice for powers of a fixed prime), it follows that M has rank $\geq r$, so K contains a torus of dimension r , say T . When H is connected of dimension r , it follows that $T = H$, which completes the proof of 8.6. Thus 8.1 is proved. $\square$

⁽⁶⁷⁾**N.D.E.** This follows, for example, from VIII 3.2 a); more generally, if $f : G \rightarrow H$ is a surjective morphism of algebraic groups over a field k and H is reduced, f is flat (cf. VI_B 1.3).

⁽⁶⁸⁾**N.D.E.** The original has been expanded in what follows.

⁽⁶⁹⁾**N.D.E.** At least in the case considered so far, namely S locally noetherian.

⁽⁷⁰⁾**N.D.E.** The editors did not understand this remark, not understanding why u would a priori have finite fibers and be surjective. . .

⁽⁷¹⁾**N.D.E.** Expand this reduction. . .

⁽⁷²⁾**N.D.E.** Expand this point. . .

Remarks 8.7. Using 8.1, it should not be difficult to give analogous generalizations of 4.7 and 4.8. A more interesting study would be that of situation 8.1 where one drops the hypothesis that the fibers of H are affine. One can show that there then exists an open neighborhood U of s such that $H|U$ is commutative and for every $t \in U$ the geometric fiber H_t is an extension of an abelian variety by a torus.⁽⁷³⁾ Of course, in questions of this kind one may restrict to the case where S is the spectrum of a discrete valuation ring, s its closed point, t its generic point.

One may generalize this result as follows. For every smooth connected algebraic group G over an algebraically closed field k , a well-known theorem of Chevalley tells us that G is (uniquely) an extension of an abelian variety A by a smooth connected affine group V . Denote by the *abelian rank* (resp. *reductive rank*, resp. *nilpotent rank*, resp. *semisimple dimension*) of G , and write $\rho_{\text{ab}}(G)$ (resp. $\rho_r(G)$, resp. $\rho_n(G)$, resp. $d_s(G)$), the dimension of A , resp. the dimension of the maximal tori of G , resp. the dimension of the Cartan subgroups⁽⁷⁴⁾ of G , resp. the dimension of the quotient of G (or again of V) by its radical (cf. *Bible* for all these notions). Introduce also the unipotent rank $\rho_u(G) = \rho_u(V) = \rho_n(G) - \rho_r(G) - \rho_{\text{ab}}(G)$. When G is not algebraically closed, we still use the same names and the same notation for the corresponding invariants of $G_{\bar{k}}$, where $\bar{k}$ is the algebraic closure of k . This being so, let G be a smooth group scheme over the spectrum S of a discrete valuation ring; let ρ_{ab} , etc. (resp. ρ'_{ab} , etc.) be the invariants associated with the special fiber (resp. generic fiber). Then one has the inequalities:

$$\begin{cases} \rho_{\text{ab}} \leq \rho'_{\text{ab}} \\ \rho_r + \rho_{\text{ab}} \leq \rho'_r + \rho'_{\text{ab}} \\ d_s \leq d'_s \end{cases} \quad \begin{cases} \rho_n \geq \rho'_n \\ \rho_u \geq \rho'_u. \end{cases}$$

It amounts to the same thing to say that if G is smooth of finite type over an arbitrary base S , the functions $s \mapsto \rho_{\text{ab}}(s)$, $\rho_{\text{ab}}(s) + \rho_r(s)$, $d_s(s)$ are lower semicontinuous, and the functions $s \mapsto \rho_n(s)$, $\rho_u(s)$ are upper semicontinuous.⁽⁷⁵⁾ The same results probably still hold without supposing G smooth over S , but merely flat of finite presentation over S , agreeing to denote, for an algebraic group G over an algebraically closed field k , by $\rho_{\text{ab}}(G)$, etc., the corresponding invariants of G_{red}^0 .

In this Seminar we present some results of this type for G affine over S , or more generally with affine fibers: in this case we shall verify the semicontinuity properties for ρ_r, ρ_n , hence for $\rho_u = \rho_n - \rho_r$, and continuity of ρ_s in a neighborhood of a point s whose fiber is a reductive group.⁽⁷⁵⁾

One may generalize 8.2 when one already supposes G commutative, as follows:

Theorem 8.8. ⁽⁷⁶⁾ *Let G be a commutative S -group prescheme flat and of finite presentation over S , with affine connected fibers. Let $s \in S$, and suppose:*

- a) If $\bar{k}$ denotes an algebraic closure of $\kappa(s)$, $(G_{\bar{k}})_{\text{red}}$ is a torus.*
- b) There exists a generization t of s such that G_t is smooth over $\kappa(t)$.*

Under these conditions there exists an open neighborhood U of s such that $G|U$ is a torus.

(Note moreover that if one merely supposes that for every generization t of s , G_t is affine resp. connected, one easily deduces that the same conclusion holds for t in an open neighborhood of s .)

⁽⁷³⁾**N.D.E.** Give a reference here?

⁽⁷⁴⁾**N.D.E.** Recall the definition of “Cartan subgroup”...

⁽⁷⁵⁾**N.D.E.** Give a reference for these results?

⁽⁷⁶⁾**N.D.E.** G. Prasad and J.-K. Yu have generalized this result (under some additional hypotheses), without supposing G commutative and replacing “torus” by “reductive group” in hypothesis a) and the conclusion, cf. [PY06], Th. 6.2.

PROOF OF 8.8. It suffices to prove that G_s is smooth over $\kappa(s)$. Indeed, as G is flat of finite presentation over S , it then follows that G is smooth over S above a neighborhood of s (cf. 3.5), but then one is under the conditions of 8.1.

To prove that G_s is smooth over $\kappa(s)$, the usual procedure reduces us to the case where S is affine noetherian. Choosing a homomorphism $S' \rightarrow S$ from the spectrum S' of a discrete valuation ring into S , sending its closed point to s and its generic point η to t , one is reduced to the case where S is itself the spectrum of a discrete valuation ring, which one may in addition suppose complete with algebraically closed residue field, and where s and $t = \eta$ are respectively the closed point and generic point of S .

Thus G is flat, separated, of finite type over S , the generic fiber G_η is smooth and connected, and the special fiber G_0 is such that $T_0 = (G_0)_{\text{red}}$ is a torus. Let m be an integer > 0 prime to the residue characteristic at s , hence also to that at η ; one then knows that $m \cdot \text{id}_G$ is an étale morphism fiber by fiber (VII_A 8.4), hence an étale morphism since G is flat over S (SGA I I 5.9); hence its kernel ${}_mG$ is étale over S , and since G is separated⁽⁷⁷⁾⁽⁷⁸⁾ of finite type over S , so is ${}_mG$. Since $({}_mG)_0 = {}_m(T_0)$, its degree is m^r , where $r = \dim T_0 = \dim G_0 = \dim G$. It follows that the rank of $({}_mG)_\eta = {}_m(G_\eta)$ is $\geq m^r$, which already proves (using 8.6) that G_η is a torus of dimension r , since the two fibers of ${}_mG$ have the same rank, and hence as in 8.5 that ${}_mG$ is finite over S .⁽⁷⁹⁾

Note that since S is complete with algebraically closed residue field, the finite étale covering ${}_mG$ splits completely, so through every point of ${}_mG_0$ passes a section of G over S ; in particular the set of points of G_0 through which passes a section of G over S is everywhere dense. Now one has the following:

Lemma 8.9. *Let S be a locally noetherian regular prescheme of dimension 1, G an S -group prescheme flat and locally of finite type such that G_η is smooth over $\kappa(\eta)$ for every maximal point η of S (which implies that G is reduced). Suppose that the normalization scheme X of G is finite over G (this is the case by Nagata if S is the spectrum of a complete discrete valuation ring, cf. EGA IV₂, 7.7.4), and let G' be the open subset of X consisting of the points at which X is smooth over S . With this notation:*

a) *If the projection $G' \rightarrow S$ is surjective, there exists on G' a unique S -group prescheme structure such that the canonical morphism $G' \rightarrow G$ is a group homomorphism.*

b) *Suppose that for every closed point s of S , the set of points of G'_s through which passes an étale quasi-section is dense in G'_s for the Zariski topology. Then X is regular, one is under the conditions of a), and the map $\Gamma(G'/S) \rightarrow \Gamma(G/S)$ is bijective.*

In the case of interest to us this lemma applies and gives a group homomorphism $u : G' \rightarrow G$, where G' is smooth over S and u_η is an isomorphism $G'_\eta \xrightarrow{\sim} G_\eta$. Replacing G' by an open subgroup if necessary, one may suppose G'_0 connected, and since $u_0 : G'_0 \rightarrow G_0$ induces a surjective morphism $G'_0 \rightarrow T_0$, where T_0 is a torus of the same dimension as G'_0 , one easily concludes that G'_0 is a torus (for example using 8.6, or referring to *Bible*, §7.3, th. 3 a)). Consequently by 8.2 G' is a torus, but then by IX 6.8, $\text{Ker } u$ is a subgroup of multiplicative type of G' , and since its generic fiber is reduced to the unit group, it is the unit group, hence u is a monomorphism. Using now VIII 7.9 it follows that u is an immersion. Being surjective, and G being reduced, it follows that u is an isomorphism, which finishes proving 8.8. $\square$

It remains to prove 8.9. To prove a), note that uniqueness of the group law on G' making $u : G' \rightarrow G$ a group homomorphism is clear, since the group law of G' is known on the generic fiber (supposing S irreducible, which is permitted). For existence, one easily reduces to the case where S is local, the spectrum of a discrete valuation ring A , and thanks to

⁽⁷⁷⁾By Raynaud's theorem VI_B 5.3.

⁽⁷⁸⁾N.D.E. Cf. N.D.E. (62) in 8.5.

⁽⁷⁹⁾N.D.E. Review the preceding sentence. . .

uniqueness and to the fact that the operation of integral closure commutes with an étale extension of the base, one may make étale extensions of the base on S , which reduces us to the case where A is “strictly local”, i.e. henselian with separably closed residue field. The same reduction applies for b), but in the hypothesis made in b) one may now replace “étale quasi-section” by “section”.

Note that G' being smooth over S and X normal, $G' \times_S X$ is normal (being smooth over normal X), so the composite morphism $G' \times_S X \longrightarrow G \times_S G \longrightarrow G$ factors as

$$p : G' \times_S X \longrightarrow X.$$

Let us prove that this latter morphism induces on the open subset $G' \times_S G'$ of $G' \times_S X$ a morphism

$$G' \times_S G' \longrightarrow G'.$$

One must therefore show that $G'_0 \times_k G'_0$ is mapped into the open subset G'_0 of X_0 ; it suffices to see that for every point g'_0 of G'_0 with values in k , the morphism

$$(x) \quad h'_0 \longmapsto p_0(g'_0, h'_0)$$

from G'_0 to X_0 takes its values in G'_0 . Now since G' is smooth over S and S henselian, every g'_0 as above is induced by a section g' of G' over S , and one verifies immediately that morphism (x) above is then induced by the morphism $h' \longmapsto p(g', h')$ from G' to X , itself deduced by transport of structure from the automorphism $h \longmapsto g \cdot h$ of G , left translation by the section g of G which is the image of g' . Thus $h' \longmapsto p(g', h')$ is itself an automorphism of X , hence maps G' to G' , which proves our assertion.

It remains to prove that the composition law so obtained on G' is a group law. Associativity follows immediately from associativity of the generic fiber (isomorphic to that of G). On the other hand, the symmetry automorphism of the S -prescheme G induces an automorphism of X , leaving G' stable and inducing an automorphism σ of G' . One then verifies that the latter has the properties of an inverse for the composition law on G' , for this again amounts to checking commutativity of certain diagrams involving fibered powers of G' over S , and these being smooth over S , it suffices to check their commutativity on the generic fiber, which is clear. This proves part a) of 8.9.

Let us prove b). Let Z' be the set of $x \in X$ such that $\mathcal{O}_{X,x}$ is not regular; it is a closed subset by a theorem of Nagata (EGA IV₂, 6.12.6), evidently contained in X_0 ; let Z be its image in G , hence a closed subset of G_0 . Then Z is a nowhere dense subset of G_0 , i.e. contains no maximal point y of G_0 . Indeed, since G_0 is defined in G by an equation $t = 0$ (where t is a uniformizer of the discrete valuation ring A), $\mathcal{O}_{G,y}$ is of dimension 1 by the Hauptidealsatz, so for every x of X over y , $\mathcal{O}_{X,x}$ is of dimension 1, hence a discrete valuation ring since X is normal and thus regular in codimension 1. On the other hand, it is evident that for every section g of G over S , Z' is stable under the automorphism of X defined by transport of structure from left translation by g in G , so Z is stable under left translation in G_0 defined by g_0 . Now by hypothesis the set of these g_0 is dense in G_0^0 . Since Z is stable under translation by these g_0 and is closed and nowhere dense, it follows immediately that $Z = \emptyset$, whence $Z' = \emptyset$, so X is regular. Consequently, X is smooth over S at every point through which passes a section. Now every section of G over S lifts uniquely to a section of X over S , and hence of G' over S . This is so in particular for the unit section, which proves that the image of G' in S is S , i.e. one is under the conditions of (a). This completes the proof of 8.9, and hence of 8.8.

9. Addenda

(80)

(80)**N.D.E.** This additional section was written by Fabrice Orgogozo (following indications of Ofer Gabber).

9.1. — Simplicial sets, topoi, groupoids and topological spaces. —

Notations 9.1.1. (1) Let E be a simplicial set. One may associate with it the following objects:

- (2) A topos $T = E^\sim$, obtained from the topoi naturally associated with the sets E_i , according to the procedure described in [Del74], 6.3.1 (see also [Ill72], VI.5.2 and SGA 4 VI.7);
- (3) A groupoid $G = \Pi(E)$ whose objects are the elements of E_0 (the “vertices”) and whose arrows are defined in [GZ67], II.7;
- (4) A topological space $X = |E|$ (a cell complex), called the “geometric” (or “topological”) “realization” (cf. loc. cit., III.1).

Let us note that a sheaf on T is nothing other than a simplicial set over E .

9.2. — Locally constant sheaves; descent data. —

Definitions 9.2.1. One calls a *locally constant sheaf* on:

- (1) A simplicial set E , every morphism of simplicial sets $E' \rightarrow E$ such that for every $i \in \mathbf{N}$ and every $e \in E_i$ the face operators d induce isomorphisms $E'_e \xrightarrow{\sim} E'_{d(e)}$ between the fibers (cf. [AM69], §10);
- (2) A topos T , every object F of T such that there exists an epimorphism $U \rightarrow 1$ and an isomorphism $F \times U \simeq f^*I \times U$, where I is a set and $f : T \rightarrow (\mathbf{Ens})$ is the final morphism (cf. SGA 4, IX.2);
- (3) A groupoid G , every presheaf on G , i.e. every contravariant functor from G to the category of sets (cf. [GZ67], append. I.1.2);
- (4) A topological space X , every sheaf of sets on X locally constant in the usual sense.

Finally:

- (5) One calls a *descent datum on a simplicial set E* the data of a sheaf F on $E_0^\sim$ (i.e. a set-theoretic function on E_0) and an isomorphism $\alpha : p_1^*F \xrightarrow{\sim} p_2^*F$, where $p_1, p_2 : E_1^\sim \rightarrow E_0^\sim$ are the morphisms (deduced from the) faces, satisfying the usual cocycle relation (cf. Exp. IV, 2.1⁽⁸¹⁾ and below).

The morphisms between these five types of objects, as well as the associated inverse image functors, are defined in the evident way.

9.3. — Some equivalences of categories. — Let E be a simplicial set, and respectively T, G, X the associated topos, groupoid and topological space.

Proposition 9.3.1. *The categories of locally constant sheaves on E, T, G, X , as well as the category of descent data on E , are equivalent.*

SKETCH OF PROOF. Denote by (1) to (5) the categories of objects defined in the preceding paragraph.

- (1) $\Leftrightarrow$ (5). This is a particular case of [AM69], 10.6 (see also [GZ67], append. I.2.3, [Fri82], ?5.6 or [Ill72], VI.8.1.6). An equivalence of categories is given by the functor associating with the object $(E' \rightarrow E)$ the pair (E'_0, α) , where E'_0 is regarded as a sheaf on E_0 , and where α is the unique isomorphism $p_1^*E'_0 \xrightarrow{\sim} p_2^*E'_0$ whose fiber at each $y \in E_1$ — with images denoted by x_1, x_2 under the two projections — is given by the isomorphisms $(E'_0)_{x_1} \xleftarrow{\sim} (E'_1)_y \xrightarrow{\sim} (E'_0)_{x_2}$.
- (5) $\Leftrightarrow$ (3). Evident: one of the two relations defining the morphisms in the groupoid G associated with E is a cocycle relation.
- (1) $\Leftrightarrow$ (4). Cf. [GZ67], append. I.3.2.1.
- (1) $\Leftrightarrow$ (2). One must show that an object over E is a locally constant sheaf in the simplicial sense if and only if it is so as a sheaf on $T = E^\sim$. This follows from the

⁽⁸¹⁾PP: I have replaced “FGA, Technique de descente I, 1.6” by “Exp. IV, 2.1”.

fact that locally constant sheaves $F_\bullet$ on T are precisely cartesian sheaves, i.e. those for which the arrows $([n] \rightarrow [m])^* F_m \rightarrow F_n$ are isomorphisms. This last point is a particular case of a general fact about simplicial topoi, together with the fact that every sheaf on $E_0^\sim$ is locally constant. (See also [Ill72], VI.8.1.6.) $\square$

For every group H , the equivalences of categories above induce equivalences between the categories of H -torsors, the latter being locally constant sheaves endowed with an action of H , with fibers isomorphic to H acting on itself by translation.

9.4. — Fundamental groups and groupoids. — It follows from the preceding equivalences of categories that for every group H and every simplicial set E , the sets of isomorphism classes of H -torsors $H^1(E, H)$, $H^1(E^\sim, H)$, $\pi_0(\text{Hom}(\Pi(E), BH))$ and $H^1(|E|, H)$ are naturally in bijection. Recall that BH denotes the one-object groupoid associated with H , and π_0 the functor associating with a category the set of isomorphism classes of its objects.

Similarly, if e is a point of E , the preceding equivalences induce bijections between the sets of isomorphism classes of H -torsors trivialized over e , denoted respectively by $H^1(E \text{ rel } e, H)$, $H^1(E^\sim \text{ rel } e^\sim, H)$, $\text{Hom}(\pi_1(\Pi(E), e), H)$ and $H^1(|E| \text{ rel } |e|, H)$. Recall that $\pi_1(\Pi(E), e)$ denotes the group $\text{Isom}_{\Pi(E)}(e, e)$. For variable H , these functors are represented, in the connected case, by a group denoted by $\pi_1(E, e)$. The group $\pi_1(E, e)$ is isomorphic to $\pi_1(\Pi(E), e)$ and $\pi_1(|E|, |e|)$, hence also to the fundamental group of a simplicial set as defined by Kan (cf. e.g. [May67], 16.1 or [Ill72], I.2.1.1). (Recall that the set $H^1(E, H)$ is for its part isomorphic to the set $H^1(\pi_1(E, e), H)$ of morphisms to H modulo conjugacy, also denoted by $\text{Hom ext}(\pi_1(E, e), H)$.)

9.5. — Cones. —

Definitions 9.5.1. (1) Let $f : E' \rightarrow E$ be a morphism of simplicial sets. Recall (cf. for example [Del74], 6.3.1) that $C(f)$ denotes the pointed simplicial set whose underlying set in degree $n \geq 0$ is

$$E_n \amalg \left(\coprod_{i < n} E'_i \right) \amalg \star,$$

where $\star$ is a singleton. We leave it to the reader to define the simplicial maps, the definition of the faces in degrees at most two being recalled below. (See also [GZ67], VI.2 for a pointed variant.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on E endowed with a trivialization of the inverse image over E' . The simplicial set $C(f)$ is naturally pointed by $\star \rightarrow C(f)$.

(2) Let $f : T' \rightarrow T$ be a morphism of topoi. Denote by $C(f)$ the topos whose objects are the quintuples $(F, F', A, \alpha : f^* F \rightarrow F', \beta : A \rightarrow \Gamma(T', F'))$, where F (resp. F') is an object of T (resp. T'), A is a set, and α (resp. β) is a morphism in T' (resp. (Ens)). (See also [Del80], 4.3.4 and [Ill72], III.4 for a variant of this construction.) The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on T endowed with a trivialization of the inverse image over T' . The topos $C(f)$ is naturally pointed by the fiber functor sending the quintuple $(F, F', A, \alpha : f^* F \rightarrow F', \beta : A \rightarrow \Gamma(T', F'))$ to the set A .

(3) Let $f : G' \rightarrow G$ be a morphism of groupoids. Denote by $C(f)$ the colimit of the diagram

$$G \leftarrow G' \rightarrow B1$$

where $B1$ is the one-object category (one object, one arrow). The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on G endowed with a trivialization of the inverse image over G' .

(4) Let $f : X' \rightarrow X$ be a morphism of topological spaces. Denote by $C(f)$ the colimit of the diagram

$$\star \leftarrow X' \times \{0\} \rightarrow X' \times [0, 1] \leftarrow X' \times \{1\} \rightarrow X.$$

The category of locally constant sheaves on $C(f)$ is equivalent to the category of locally constant sheaves on X endowed with a trivialization of the inverse image over X' . The topological space $C(f)$ is naturally pointed by $\star \rightarrow C(f)$.

9.5.2. — Descent data on the cone of a simplicial map. Let $f : E' \rightarrow E$ be a morphism of simplicial sets and $C(f)$ its cone (cf. above, (1)). Use the letter p (resp. q, r) to denote the face maps of E' (resp. $E, C(f)$). Before stating the proposition below, let us make explicit the faces r in degrees at most two in terms of p, q . By convention, p_{ij} (resp. p_i) is the face map $E'_2 = E'_{\{1,2,3\}} \rightarrow E'_1 = E'_{\{1,2\}}$ (resp. $E'_1 = E'_{\{1,2\}} \rightarrow E'_0 = E'_{\{1\}}$) corresponding to the increasing map $\{1, 2\} \rightarrow \{1, 2, 3\}$ (resp. $\{1\} \rightarrow \{1, 2\}$) with image $\{i, j\}$ (resp. $\{i\}$).⁽⁸²⁾ One adopts the same convention for the faces of E and $C(f)$. The morphism

$$r_1 : C(f)_1 = E_1 \coprod E'_0 \coprod \star \rightarrow C(f)_0 = E_0 \coprod \star$$

(resp. r_2) is:

- $q_1 : E_1 \rightarrow E_0$ (resp. $q_2 : E_1 \rightarrow E_0$) on E_1 ;
- $E'_0 \rightarrow \star$ (resp. $f_0 : E'_0 \rightarrow E_0$) on E'_0 ;
- $\star \rightarrow \star$ on $\star$.

Similarly, the morphism

$$r_{21} : C(f)_2 = E_2 \coprod E'_1 \coprod E'_0 \coprod \star \rightarrow C(f)_1 = E_1 \coprod E'_0 \coprod \star$$

(resp. r_{32}, r_{31}) is:

- $q_{21} : E_2 \rightarrow E_1$ (resp. q_{32}, q_{31}) on E_2 ;
- $p_1 : E'_1 \rightarrow E'_0$ (resp. $f_1 : E'_1 \rightarrow E_1, p_2 : E'_1 \rightarrow E'_0$) on E'_1 ;
- $E'_0 \rightarrow \star$ (resp. $\text{Id} : E'_0 \rightarrow E'_0, \text{Id} : E'_0 \rightarrow E'_0$) on E'_0 .

Let H be a descent datum on $C(f)$, i.e. a sheaf H_0 on $C(f)_0 = E_0 \coprod \star$ endowed with an isomorphism $\gamma : r_1^* H_0 \xrightarrow{\sim} r_2^* H_0$ between its inverse images on $C(f)_1 = E_1 \coprod E'_0 \coprod \star$ satisfying the cocycle relation $r_{31}^* \gamma = r_{32}^* \gamma \circ r_{21}^* \gamma$.

The restriction of the isomorphism γ to E_1 (resp. E'_0) is a descent datum $\beta : q_1^* G_0 \xrightarrow{\sim} q_2^* G_0$, where G_0 is the restriction of H_0 to E_0 (resp. a trivialization $t : \underline{H_0(\star)} \xrightarrow{\sim} f^* G_0 =: F_0$, where $\underline{H_0(\star)}$ is the constant sheaf with stalk $H_0(\star)$). The restriction of the cocycle relation satisfied by γ to E_2 (resp. E'_1) is the cocycle relation for β (resp. $p_2^*(t) = f_1^*(\beta) \circ p_1^*(t)$). This latter relation means that the trivialization t is compatible with the descent datum $f_1^*(\beta)$ induced by β on F_0 . It follows that:

Proposition 9.5.3. *Let $f : E' \rightarrow E$ be a morphism of simplicial sets and $C(f)$ its cone, pointed by $\star$. The category of descent data on $C(f)$ trivialized over $\star$ is equivalent to the category of descent data on E endowed with a trivialization of the descent datum induced on E' .*

9.6. — Representability of the functor $\pi^1(S'/S, \xi; -)$. — The notation is that of page 90, and one henceforth supposes the connected components of $S_0 = S'$ and $S_1 = S' \times_S S'$ open. Under this hypothesis one has the following explicit combinatorial description, which follows immediately from the recollections above.

Proposition 9.6.1. *For every group G , the set $\pi^1(S'/S, \xi; G)$ of isomorphism classes of descent data endowed with a trivialization over ξ is in natural bijection with the relative nonabelian cohomology set $H^1(K \text{ rel } k, G) := H^1(\tilde{K} \text{ rel } \star, G)$.*

It follows that this functor is representable by a group, denoted by $\pi_1(K, k)$ (the “relative fundamental group”), as soon as the simplicial set $\tilde{K} = C(k \rightarrow K)$ is connected. One verifies immediately that this is so if and only if the map $\pi_0(k) \rightarrow \pi_0(K)$ is surjective. In particular

⁽⁸²⁾This convention departs from the current simplicial standard but is closer to the notation used by Grothendieck in his descent theory.

it suffices that the simplicial set K be connected. As indicated in the text, this is the case if the scheme S is connected and the morphism $S' \rightarrow S$ universally submersive.

9.7. — Contractibility of k , counterexample to injectivity of $\mathbf{k}_\bullet \rightarrow \mathbf{K}_\bullet$. — The following two propositions make N.D.E. (53) more precise.

Proposition 9.7.1. *Let κ be an algebraically closed field, and X a connected κ -scheme. For every integer $i \geq 0$, denote by X_i the $(i+1)$ -st fibered power of X over κ . For every integer $i \geq 0$, the canonical map*

$$\pi_0(X_i) \longrightarrow \pi_0(X)^{i+1}$$

is a bijection. In particular, the simplicial set $\pi_0(X_\bullet)$ is contractible.

It suffices to prove the following lemma, which follows by passage to the limit from the Künneth formula. [Expand?]

Lemma 9.7.2. *Let κ be an algebraically closed field and X, Y two connected κ -schemes. Then the fibered product $X \times_\kappa Y$ is connected.*

Proposition 9.7.3. *Let X be a connected scheme, $X' \rightarrow X$ a finite étale morphism and $\bar{x}$ a geometric point of X located at a point x . The canonical morphism $\pi_0(X'_\bar{x}) \rightarrow \pi_0(X')$ is surjective.*

Indeed, every connected component of X' has as image an open and closed subset of X , and hence meets the fiber X'_x . The morphism $\pi_0(X'_x) \rightarrow \pi_0(X')$ is therefore surjective, as is the morphism $\pi_0(X'_\bar{x}) \rightarrow \pi_0(X'_x)$.

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⁽⁸³⁾**N.D.E.** Additional references cited in this Exposé; references [AM69] and following concern the Addenda.